Suboptimal Frequentist Performance of the Flat-Prior Generalized Bayes Optimal Algorithm in Best Arm Identification

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Abstract

We consider the fixed-budget best arm identification problem with rewards following normal distributions. In this problem, the forecaster is given K arms (or treatments) and T time steps. The forecaster attempts to find the arm with the largest mean, via an adaptive experiment conducted using an algorithm. The algorithm's performance is evaluated by "simple regret", reflecting the quality of the estimated best arm. While frequentist simple regret can decrease exponentially with respect to T, the optimal Bayesian simple regret admits an $O(1/T)$ upper bound under suitable regularity conditions on the prior. This paper demonstrates that the flat-prior generalized Bayes algorithm, which minimizes posterior expected simple regret after a fixed one-sample-per-arm initialization, does not yield an exponential decrease in simple regret under certain parameter settings. This contrasts with the numerous findings that suggest the asymptotic equivalence of Bayesian and frequentist approaches in fixed sampling regimes. Although this generalized Bayes optimal algorithm is formulated by a recursive equation that is virtually impossible to compute exactly, we lay the groundwork for its analysis by introducing a novel concept termed the *expected Bellman improvement*.

Quote

The following paragraph appears in a conversation between Joseph Leo Doob and James Laurie Snell [Snell, 1997].

> While writing my book I (Doob) had an argument with Feller. He asserted that everyone said "random variable" and I asserted that everyone said "chance variable." We obviously had to use the same name in our books, so we decided the issue by a stochastic procedure. That is, we tossed for it and he won.

The coin that Doob and Feller tossed separated our world, where we discuss the randomness of random variables, from a parallel world where we discuss the chanciness of chance variables. In this paper, we consider a generalized Bayes optimal algorithm, which computes the consequences over all the exponentially many parallel worlds.

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1 Introduction

This paper addresses the problem of identifying the best arm (or treatment) among multiple options from a fixed number of samples. This problem associates each arm with an (unknown) parameterized distribution, which is a (noisy) signal of the quality of the treatment. To maximize the overall effectiveness of the treatments, the forecaster uses an algorithm that adaptively determines which arms to select. Such adaptive allocation is essential for solving many practical problems, including A/B testing in online platforms [Li et al., 2010], simulation optimization [Malkomes et al., 2021], and clinical trials [Thompson, 1933, Villar et al., 2015]. Among the best-known algorithms for sequential selection are the multi-armed bandit algorithms, such as Thompson sampling [Thompson, 1933], and upper confidence bounds [Lai and Robbins, 1985, Auer et al., 2002]. Although multi-armed bandit algorithms solve the exploration–exploitation tradeoff to maximize total effectiveness during the rounds, they have limited ability to detect the best arm [Bubeck et al., 2009], and are thus suboptimal for our aim.

The machine learning literature refers to the process of finding the best-quality arm as best arm identification (BAI; Audibert et al. 2010). Unlike multi-armed bandit algorithms, BAI algorithms are designed solely to deliver the most effective exploration.[1] BAI has two settings, namely, the fixed-budget setting which the number of samples is fixed, and the fixed-confidence setting in which the forecaster stops when the confidence level on the best arm reaches a predefined threshold. In this paper, we consider the former setting. Popular algorithms for fixed-budget identification include successive rejects (SR; Audibert et al. 2010) and successive halving [Karnin et al., 2013]. There are also several Bayesian algorithms that utilize a prior, such as top-two Thompson sampling [Russo, 2020], knowledge gradient (KG; Gupta and Miescke 1994), and expected improvement (EI; Jones et al. 1998). A Bayesian algorithm is initialized with a prior belief, and the forecaster learns from each sample to build a posterior belief. Given that, we know the dynamics of the posterior belief as a distribution, exact minimization of the Bayes risk can be formulated as dynamic programming. Nonetheless, exact dynamic programming solutions are computationally intractable because of the curse of dimensionality [Russo, 2020, Powell and Ryzhov, 2013]. Consequently, KG and EI are designed to provide the one-step lookahead approximations of an exactly optimal dynamic programming solution.

While some classes of Bayesian algorithms are known to perform optimally in the fixed-confidence setting [Shang et al., 2020], limited results are known regarding their performance in the fixed-budget setting. This paper considers the flat-prior generalized Bayes optimal algorithm in the context of fixed-budget identification. In the field of statistical decision theory, the Bayes optimal algorithm is deeply connected to a minimax estimator that maximizes the worst-case performance. We are interested in the following research question: Does this generalized Bayes optimal algorithm have a good performance in the frequentist measure? On the contrary, we show that the flat-prior generalized Bayes optimal algorithm performs sub-optimally with some of the worst model parameters, which implies that maximizing the posterior objective differs substantially from maximizing the frequentist objective. A Bayesian measure requires it to optimize the performance averaged over the prior, whereas a frequentist measure requires

[1] Although the term "best arm identification" has appeared only recently, several strands of research share the same goal, among which ranking and selection (RS; Bechhofer 1954) is among the best known. Several works, such as Russo [2020], Hong et al. [2021], consider the difference between BAI and RS, and we briefly discuss work related to RS in Section 6. However, because BAI and RS share several overlapping ideas, we do not explicitly outline the difference.

an algorithm to optimize the performance against any case. The fact that the generalized Bayes optimal algorithm is suboptimal means that even if we enhance KG and EI to plan more than one step ahead, their performance in a frequentist measure might not improve.

Note that this discrepancy between Bayesian and frequentist measures differs considerably from the situation in standard statistical inference with non-adaptive samples. For a fixed-sample problem, the Bernstein–von Mises theorem describes the asymptotic equivalence of Bayesian and frequentist inference. However, in an adaptive sampling scheme, underestimation of an arm due to the randomness of the empirical mean results in a smaller number of samples of the arm in the future. Bayesian and frequentist BAI algorithms are both robust against such randomness but with different confidence levels. Bayesian algorithms are robust up to a polynomially small underestimation, whereas frequentist algorithms are robust up to an exponentially small underestimation.

Our results offer analytical innovation by establishing foundational principles for a formal analysis that yields exact solutions in dynamic programming. This is notable because it enables solutions that extend beyond the conventional one- or two-step lookahead. We demonstrate several instances in which it is feasible to perform exact analyses of dynamic programming regardless of the need to project the evolution of the posterior over extended future periods

The structure of the paper is as follows. The rest of this section establishes the problem. Section 2 formalizes the Bayes optimal algorithm. Section 3 describes our main results, and Sections 4–5 describes their derivation. Section 6 discusses related work, and Section 7 concludes the paper. Most details of the proofs appear in the appendices.

1.1 Problem setup

Fix integers $K \geq 2$ and $T \geq K$. We consider the normal BAI problem with K arms and T samples. Each arm $i \in [K] = \{1, 2, \ldots, K\}$ is associated with an (unknown) parameter $\mu_i \in \mathbb{R}$. We use $\boldsymbol{\mu} = (\mu_1, \mu_2, \ldots, \mu_K)$ to denote the set of parameters associated with the arms. Under a fixed parameter $\boldsymbol{\mu}$, let

$$\big(X_{i,n}\big)_{i\in[K],\, n\geq 1}$$

be mutually independent random variables such that $X_{i,n} \sim \mathcal{N}(\mu_i, 1)$. The sequence $(X_{i,n})_{n\geq 1}$ is the reward stack of arm i. The problem involves T rounds, and at each round $t = 1, 2, \ldots, T$, the forecaster can choose one of the arms. If arm i has previously been chosen $n-1$ times, choosing it at round t reveals $X(t) := X_{i,n}$. In particular, conditionally on the past and on $I(t) = i$, $X(t) \sim \mathcal{N}(\mu_i, 1)$. Here, $\mathcal{N}(\mu_i, 1)$ is a normal distribution with mean μ_i and unit variance.

Put $\mu^* := \max_{i\in[K]} \mu_i$ and fix the tie-breaking convention $i^* := \min \arg\max_{i\in[K]} \mu_i$. The goal of the forecaster is to identify an arm with mean μ^* using an effective algorithm. Any such algorithm comprises a sampling rule $I(1), I(2), I(3), \ldots, I(T)$ and a recommendation rule $J(T)$. Let $U(0), U(1), \ldots$ be i.i.d. standard normal external randomizers, independent of all reward stacks, and define

$$\mathcal{F}_t := \sigma(U(0), \ldots, U(t), I(1), X(1), \ldots, I(t), X(t))\,, \qquad t = 0, 1, \ldots, T,$$

where the lists involving $I(s), X(s)$ are empty when $t = 0$. Then the sampling rule $I(t)$ is $\mathcal{F}_{t-1}$-measurable, meaning that the algorithm chooses an arm at round t according to the observation history $\{I(s), X(s)\}_{s<t}$

and the external randomizers $\{U(s)\}_{s<t}$.[2] An algorithm is deterministic when it does not use the information of $\{U(s)\}_{s<t}$.

Formally, an algorithm is fixed independently of the unknown parameter $\boldsymbol{\mu}$. Thus, the same Borel decision maps are used under every $\mathbb{P}_{\boldsymbol{\mu}}$: for suitable Borel maps φ_t and ψ_T,

$$I(t) = \varphi_t\big(U(0{:}t-1), I(1{:}t-1), X(1{:}t-1)\big),$$
$$J(T) = \psi_T\big(U(0{:}T), I(1{:}T), X(1{:}T)\big).$$

The parameter $\boldsymbol{\mu}$ changes only the law of the reward stacks, not these decision maps.

The recommendation rule is used to choose an arm $J(T) \in [K]$ that is $\mathcal{F}_T$-measurable. The performance of the forecaster, which follows algorithm A, is measured by the simple regret

$$\mathrm{R}_{\boldsymbol{\mu}}(T, A) = \mu^* - \mathbb{E}_{\boldsymbol{\mu}}[\mu_{J(T)}], \tag{1}$$

which is the expected difference between the means of the best arm and the recommended arm $J(T)$. We omit A from $\mathrm{R}_{\boldsymbol{\mu}}(T, A)$ and $\mathrm{R}_{\boldsymbol{H}}(T, A)$ when it is clear which algorithm is used.

The simple regret formulated by Eq. (1) is frequentist in the sense that it assumes that $\boldsymbol{\mu}$ is fixed (but unknown). In contrast, a Bayesian approach considers a known proper distribution $\boldsymbol{H}$ of $\boldsymbol{\mu} \in \mathbb{R}^K$ and evaluates an algorithm with respect to these expected values. When using the ordinary Bayesian risk below, we assume $\mathbb{E}_{\boldsymbol{H}}[\max_i |\mu_i|] < \infty$. More concretely, the Bayesian simple regret is given by

$$\mathrm{R}_{\boldsymbol{H}}(T, A) = \mathbb{E}_{\boldsymbol{H}}[\mathrm{R}_{\boldsymbol{\mu}}(T, A)] := \int \mathrm{R}_{\boldsymbol{\mu}}(T, A) d\boldsymbol{H}(\boldsymbol{\mu}),$$

which marginalizes $\mathrm{R}_{\boldsymbol{\mu}}(T, A)$ over the prior $\boldsymbol{H} = \boldsymbol{H}(\boldsymbol{\mu})$.

1.2 Convergence rates: Frequentist simple regret compared with Bayesian simple regret

A good horizon-indexed family of algorithms, from the frequentist perspective, achieves exponentially small regret as a function of T at every instance with a unique best arm.

Definition 1. (Frequentist rate) Let $(A_T)_{T \geq K}$ be a horizon-indexed family of BAI algorithms, and let Γ satisfy $0 < \Gamma(\boldsymbol{\mu}) < \infty$ for every $\boldsymbol{\mu}$ with a unique best arm. The family $(A_T)_{T \geq K}$ has frequentist rate $\Gamma(\boldsymbol{\mu})$ if, for every such $\boldsymbol{\mu}$,

$$\mathrm{R}_{\boldsymbol{\mu}}(T, A_T) \leq e^{-\Gamma(\boldsymbol{\mu})T + o(T)},$$

where $o(T)$ is a function of $K, \boldsymbol{\mu}$, and T that grows more slowly than T when we view K and $\boldsymbol{\mu}$ as constants.

Put

$$H_1(\boldsymbol{\mu}) := \sum_{i \neq i^*} \frac{1}{(\mu^* - \mu_i)^2}.$$

[2]

There is a universal constant $c_{\mathrm{SR}} > 0$ such that the successive rejects algorithm [Audibert et al., 2010] has the frequentist rate $\Gamma_{\mathrm{SR}}(\boldsymbol{\mu}) = c_{\mathrm{SR}}/(\log K\, H_1(\boldsymbol{\mu}))$. For unit-variance Gaussian arms, the uniform algorithm that draws every arm equally and recommends the largest empirical mean has the rate

$$\Gamma_{\mathrm{unif}}(\boldsymbol{\mu}) = \frac{1}{4K} \min_{i \neq i^*} (\mu^* - \mu_i)^2,$$

up to the rounding of the allocation. Depending on the instance, this rate can be a factor of order K, up to logarithmic terms, smaller than the rate of SR. More recently, Komiyama et al. [2022] derived the possible value of the rate $\Gamma(\boldsymbol{\mu})$.

For the regular prior class considered by Komiyama et al. [2021], the optimal Bayesian simple regret admits an $O(1/T)$ upper bound. More concretely, they showed that

$$\mathrm{R}_{\boldsymbol{H}}(T) \leq \frac{C_{\mathrm{bayes}}(\boldsymbol{H})}{T} + o\left(\frac{1}{T}\right),$$

where the value $C_{\mathrm{bayes}}(\boldsymbol{H})$ depends on the prior $\boldsymbol{H}$.

Remark 1. (Reward distribution) The upper-bound analysis of Audibert et al. [2010] assumes rewards supported on $[0, 1]$, whereas the lower-bound construction of Carpentier and Locatelli [2016] uses Bernoulli arms. Corresponding exponential guarantees can also be derived for Gaussian arms with unit variance. Indeed, a Gaussian empirical mean satisfies a Gaussian concentration inequality with the same exponential form as Hoeffding's inequality; this supplies the concentration step needed in the analogous upper-bound argument. The corresponding Gaussian lower bounds follow by the same change-of-measure method using Gaussian Kullback–Leibler divergences. Elsewhere, while the results of Komiyama et al. [2021] apply specifically to Bernoulli arms, an analogous $O(1/T)$ upper bound can be derived for Gaussian arms under corresponding regularity assumptions, provided we are not interested in the magnitude of the constant.

This paper considers the flat-prior generalized Bayes algorithm defined below. The algorithm draws every arm once and, conditional on the resulting proper posterior, minimizes posterior expected simple regret at every subsequent state. Our main result demonstrates that the flat-prior generalized Bayes algorithm does not feature an exponential frequentist simple regret, despite its pointwise posterior optimality after initialization.

2 Flat-Prior Generalized Bayes Optimal Algorithm and Loss Function

In this section, we formalize the flat-prior generalized Bayes algorithm that minimizes posterior expected simple regret after the fixed initialization.

2.1 Flat-prior generalized Bayes optimal algorithm

First, we define the flat-prior generalized Bayes optimal algorithm, and then we represent it as a dynamic programming problem.

Definition 2. (Algorithms and posterior-state policies) An algorithm A is

$$A = (I(1), \dots, I(T), J(T)),$$

where each $I(t)$ is $\mathcal{F}_{t-1}$-measurable and $J(T)$ is $\mathcal{F}_T$-measurable. Let $\mathcal{A}$ be the set of all such algorithms, and put

$$\mathcal{A}_{\text{init}} := \{A \in \mathcal{A} : I(t) = t \text{ for every } t \in [K]\}.$$

Thus every algorithm in $\mathcal{A}_{\text{init}}$ samples each arm once during the fixed initialization.

Let

$$\mathcal{G}_K := \left\{ \bigotimes_{k=1}^{K} \mathcal{N}(m_k, 1/n_k) : m_k \in \mathbb{R},\ n_k > 0 \right\}$$

be the proper product-Gaussian posterior state space. We identify $\mathcal{G}_K$ with $\mathbb{R}^K \times (0,\infty)^K$ through $(m_1, \dots, m_K, n_1, \dots, n_K)$ and equip it with the corresponding Borel σ-field. A horizon-T posterior-state policy is a collection $q = (q_{K+1}, \dots, q_{T+1})$ of Borel maps $q_t : \mathcal{G}_K \times \mathbb{R} \to [K]$. Following the fixed initialization, it induces a history algorithm $A^q \in \mathcal{A}_{\text{init}}$ by

$$\begin{aligned} I(t) &= q_t\big(\boldsymbol{H}(t-1), U(t-1)\big), && t = K+1, \dots, T, \\ J(T) &= q_{T+1}\big(\boldsymbol{H}(T), U(T)\big). \end{aligned}$$

The second argument realizes randomized kernels; deterministic policies simply ignore it.

Definition 3. (Flat-prior generalized Bayes optimal algorithm) Equip each μ_i with Lebesgue measure, independently across arms, and use the fixed initialization. The resulting posterior $\boldsymbol{H}(K)$ belongs to $\mathcal{G}_K$ for every realized initialization.

For $t \in \{K+1, \dots, T+1\}$ and $\boldsymbol{H} \in \mathcal{G}_K$, let $\Pi_t(\boldsymbol{H})$ be the class of continuation policies satisfying the same nonanticipativity and parameter-independence requirements as the algorithms above. Concretely, adjoin i.i.d. standard normal randomizers $V(t-1), \dots, V(T)$, independent of $\boldsymbol{\mu}$ and all continuation rewards. Every $\pi \in \Pi_t(\boldsymbol{H})$ is represented by Borel maps a_s^π and d^π such that

$$\begin{aligned} I^\pi(s) &= a_s^\pi\big(\boldsymbol{H}, V(t-1{:}s-1), I^\pi(t{:}s-1), X^\pi(t{:}s-1)\big), && s = t, \dots, T, \\ J^\pi(T) &= d^\pi\big(\boldsymbol{H}, V(t-1{:}T), I^\pi(t{:}T), X^\pi(t{:}T)\big), \end{aligned}$$

where empty index ranges are omitted. These decision maps are fixed before $\boldsymbol{\mu} \sim \boldsymbol{H}$ is drawn and are the same for every realized value of $\boldsymbol{\mu}$. At each sampling round they use only the displayed, previously revealed information; in particular, no map has access to the latent parameter or to an unobserved reward. Thus a continuation policy may be history-dependent and randomized, but is nonanticipative and independent of the latent parameter. When $t = T+1$, only the terminal recommendation remains. For $\pi \in \Pi_t(\boldsymbol{H})$, let $\mathbb{E}^\pi_{\boldsymbol{H}}$ denote expectation under the joint continuation law obtained by drawing $\boldsymbol{\mu} \sim \boldsymbol{H}$, generating future rewards conditionally on $\boldsymbol{\mu}$, and applying π, including its external randomization.

For a posterior-state policy q, write $\pi^q_{t,T}(\boldsymbol{H})$ for the continuation induced by $q_t, \dots, q_{T+1}$ from state $(\boldsymbol{H}, t)$. The policy q^* is *sequentially posterior optimal* if, for every $t \in \{K+1, \dots, T+1\}$ and every

$\boldsymbol{H} \in \mathcal{G}_K$,

$$\pi^{q^*}_{t,T}(\boldsymbol{H}) \in \operatorname*{arg\,min}_{\pi \in \Pi_t(\boldsymbol{H})} \mathbb{E}^{\pi}_{\boldsymbol{H}} \left[\mu^* - \mu_{J^\pi(T)}\right]. \tag{2}$$

A flat-prior generalized Bayes optimal algorithm A^*_T is any algorithm A^{q^*} induced by a sequentially posterior-optimal state policy q^*.

In particular, when $T > K$, every $\boldsymbol{H} \in \mathcal{G}_K$ satisfies

$$\pi^{q^*}_{K+1,T}(\boldsymbol{H}) \in \operatorname*{arg\,min}_{\pi \in \Pi_{K+1}(\boldsymbol{H})} \mathbb{E}^{\pi}_{\boldsymbol{H}}[\mu^* - \mu_{J^\pi(T)}]. \tag{3}$$

When $T = K$, Eq. (2) at $t = T + 1$ gives the terminal posterior Bayes recommendation directly after initialization.

The improper measure is used only to obtain the posterior after the common fixed initialization; we never integrate a risk with respect to that measure. Thus, Definition 3 is a pointwise posterior (generalized Bayes) definition. The pointwise requirement makes behavior at arbitrary posterior states explicit but does not impose a canonical tie-breaking rule among posterior-optimal actions. For randomized state policies, statements below that an action or recommendation belongs to a specified set are understood pointwise in the posterior state and almost surely in the corresponding external randomizer; modifications on randomizer-null sets are immaterial. This posterior-state convention identifies histories having the same posterior. It does not change risk relative to optimization only at $\boldsymbol{H}(K)$. Indeed, off-path actions can be replaced without changing the on-path law, and the Bellman principle implies that a policy satisfying Eq. (3) can choose a nonminimizing action only on a set of posterior-predictive histories of probability zero; otherwise a local replacement by a minimizing continuation would strictly reduce its posterior risk. Over a finite horizon, both the posterior-predictive law and the law under every fixed Gaussian parameter $\boldsymbol{\mu}$ have everywhere-positive densities for the observed reward vector. They are therefore mutually absolutely continuous. Such null-history modifications also have zero frequentist probability and do not change $\mathrm{R}_{\boldsymbol{\mu}}(T)$. We hence identify policies that differ only by these null-history modifications with their sequentially optimal versions.

The same Gaussian-state dynamic program applies whenever the posterior at the start of continuation is a proper product-Gaussian distribution; this includes independent proper Gaussian priors as well as the initialized flat prior used here. For a general proper prior with finite first moment, the Bellman principle still applies on the full posterior-distribution state space, but the Gaussian sufficient-statistic recursion below need not apply. Unless the reduced two-armed problem is explicitly under discussion, "Bayes optimal algorithm" below refers to the flat-prior generalized Bayes optimal algorithm.

2.2 Posterior and sufficient statistics

We use the flat measure from Definition 3, which results in the posterior mean matching the empirical mean after initialization. By definition, $T \geq K$ and $I(t) = t$ for $t \in [K]$, so every posterior used by the continuation policy is proper.

For $t \geq K$, the sufficient statistics $(S_i(t), N_i(t))$ for each arm i are

$$S_i(t) = \sum_{s \leq t: I(s)=i} X(s)$$
$$N_i(t) = \sum_{s \leq t: I(s)=i} 1,$$

and the posterior of arm i at the end of round $t \geq K$ is identified by these sufficient statistics $\boldsymbol{H}(t) = (H_1(t), H_2(t), \ldots, H_K(t))$ as

$$H_i(t) = \mathcal{N}\left(\frac{S_i(t)}{N_i(t)}, \frac{1}{N_i(t)}\right).$$

The posterior on $\boldsymbol{\mu}$ is fully specified by the sufficient statistics. Henceforth, we use the posterior $\boldsymbol{H}(t)$ and the sufficient statistics $(S_i(t), N_i(t))_{i \in [K]}$ interchangeably. We also recognize that $\hat{\mu}_i(t) = S_i(t)/N_i(t)$ and $(\hat{\mu}_i(t), N_i(t))_{i \in [K]}$ represent sufficient statistics.

2.3 State evolution

The computation of the Bayes optimal algorithm can be formalized as an undiscounted episodic Markov decision process (MDP). In particular, the state at the current round t is represented as a set of sufficient statistics $\boldsymbol{H}^{\mathrm{r}}(t-1) = \boldsymbol{H}(t-1)$. We use $X^{\mathrm{r}}(\cdot)$, $N^{\mathrm{r}}(\cdot)$, $\hat{\mu}_i^{\mathrm{r}}(\cdot)$, $I^{\mathrm{r}}(\cdot)$, $J^{\mathrm{r}}(\cdot)$, and $\boldsymbol{H}^{\mathrm{r}}(\cdot)$ to represent the corresponding quantity in the posterior.[3] The state transition is represented as follows: when we draw arm i, $H_j^{\mathrm{r}}(t) = H_j^{\mathrm{r}}(t-1)$ for $j \neq i$. For $H_i^{\mathrm{r}}(t)$,

$$\begin{aligned}
\tilde{\mu}_i &\sim \mathcal{N}\left(\hat{\mu}_i^{\mathrm{r}}(t-1), \frac{1}{N_i^{\mathrm{r}}(t-1)}\right) \\
X^{\mathrm{r}}(t) &\sim \mathcal{N}(\tilde{\mu}_i, 1) \\
\hat{\mu}_i^{\mathrm{r}}(t) &= \frac{N_i^{\mathrm{r}}(t-1)\hat{\mu}_i^{\mathrm{r}}(t-1) + X^{\mathrm{r}}(t)}{N_i^{\mathrm{r}}(t-1) + 1} \\
N_i^{\mathrm{r}}(t) &= N_i^{\mathrm{r}}(t-1) + 1.
\end{aligned}$$

In view of the MDP, the algorithm receives no loss[4] for choosing an action because this is a pure exploration setting.

For later use, we define the transition on an arbitrary proper product-Gaussian posterior. Write

$$\boldsymbol{H} = \bigotimes_{k=1}^{K} \mathcal{N}(m_k, 1/n_k), \qquad n_k > 0,$$

and let $\mathsf{U}_i(\boldsymbol{H}, x)$ be the posterior obtained after observing reward x from arm i: its i-th mean and

[3] At any given time step t, an algorithm can only access the sequences $I(1), X(1), I(2), X(2), \ldots, I(t-1), X(t-1)$ that have been drawn from the true distribution. The Bayes optimal algorithm at round t calculates the value function and the action value function using a recursive formula that includes samples from the future time steps $t, t+1, t+2, \ldots, T$. These are virtual samples, drawn from the posterior distribution.

[4] This is because the values $X(1), X(2) \ldots, X(T)$ do not directly affect the final objective of minimizing the simple regret. We use the term "reward" for the values $X(1), X(2) \ldots, X(T)$, whereas we use the terms "loss" or "simple regret" for representing the quantity that we would like to minimize.

precision are $(n_i m_i + x)/(n_i + 1)$ and $n_i + 1$, respectively, and all other coordinates are unchanged. The posterior-predictive law of that reward is

$$X_i \mid \boldsymbol{H} \sim \mathcal{N}\left(m_i, 1 + \frac{1}{n_i}\right). \tag{4}$$

For any proper product-Gaussian posterior $\boldsymbol{G}$, let $\mathsf{m}_j(\boldsymbol{G})$ denote its j-th posterior mean; in particular, $\mathsf{m}_j(\boldsymbol{H}) = m_j$ for the representation above. We call a posterior reachable if it is obtained from the fixed one-sample-per-arm initialization by finitely many Gaussian updates, or by restricting such a posterior to an active subset of arms. Every reachable posterior has positive-integer precisions n_k. All actual and posterior-predictive states used in the proof of Theorem 2 are reachable in this sense.

2.4 Value function and action value function

The second argument of the loss function records the number of remaining sampling opportunities, not the number of observations used to construct its first argument. Thus, for every proper product-Gaussian posterior $\boldsymbol{H}$, we define $L(\boldsymbol{H}, t)$ for $t \in \{K+1, \dots, T+1\}$, even when $\boldsymbol{H}$ was obtained after a number of observations different from $t-1$. This convention makes the skipped-round value $L(\boldsymbol{H}(t-1), t+1)$ well-defined.

Fix the measurable terminal tie-breaking rule

$$\widehat{J}(\boldsymbol{H}) := \min \arg\max_{k \in [K]} m_k. \tag{5}$$

The terminal posterior loss is

$$L(\boldsymbol{H}, T+1) := \mathbb{E}_{\boldsymbol{H}}\left[\mu^* - \mu_{\widehat{J}(\boldsymbol{H})}\right]. \tag{6}$$

For $t = T, T-1, \dots, K+1$, define the action loss and value function by the backward recursion

$$L_i(\boldsymbol{H}, t) := \mathbb{E}_{X_i \mid \boldsymbol{H}}\left[L\big(\mathsf{U}_i(\boldsymbol{H}, X_i), t+1\big)\right], \tag{7}$$

$$L(\boldsymbol{H}, t) := \min_{i \in [K]} L_i(\boldsymbol{H}, t). \tag{8}$$

For any nonempty active arm set $\mathcal{I} \subseteq [K]$, write $\boldsymbol{H}_{\mathcal{I}}$ for the corresponding product posterior and put

$$\mu^*_{\mathcal{I}} := \max_{k \in \mathcal{I}} \mu_k, \qquad \Delta^{\mathcal{I}}_j := \mu^*_{\mathcal{I}} - \mu_j, \qquad \widehat{J}_{\mathcal{I}}(\boldsymbol{H}_{\mathcal{I}}) := \min \arg\max_{j \in \mathcal{I}} m_j.$$

Define $L^{\mathcal{I}}$ and $L^{\mathcal{I}}_i$ by Eqs. (6)–(8), with $[K]$ replaced by $\mathcal{I}$, μ^* by $\mu^*_{\mathcal{I}}$, and $\widehat{J}$ by $\widehat{J}_{\mathcal{I}}$. The stage domains are unchanged by this restriction: $L^{\mathcal{I}}(\cdot, t)$ is defined for $t \in \{K+1, \dots, T+1\}$, whereas $L^{\mathcal{I}}_i(\cdot, t)$ is defined for $t \in \{K+1, \dots, T\}$. We suppress the superscript for the full problem. We also suppress it in the reduced two-armed problem; thus, whenever the argument is $\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}$, every maximum, recommendation, loss, and gap Δ_j is restricted to arms 1 and 2. Equation (7) is equivalent to integrating first over $\tilde{\mu}_i \sim \mathcal{N}(m_i, 1/n_i)$ and then over $X_i \sim \mathcal{N}(\tilde{\mu}_i, 1)$, as in the state evolution above.

All these losses are finite because Gaussian first moments are finite. They are Borel measurable by backward induction. Since the action set is finite, the argmin is nonempty at every state, and choosing its

smallest element gives a measurable deterministic Markov policy. Randomized Bayes optimal policies may randomize only among the minimizing actions, almost surely with respect to their external randomizers.

2.5 Expected Bellman improvement

The loss function $L(\boldsymbol{H}(t-1), t)$ is recursively defined and "virtually impossible to solve" [Powell and Ryzhov, 2013, Russo, 2020]. However, we derive a lower bound for a specific example by introducing several novel ideas. First, leveraging the fact that the only actual loss is incurred at the terminal state, we define the expected Bellman improvement (EBI) for an arm, which represents the expected value gained from sampling that arm.

Definition 4. (Expected Bellman improvement of arm i) For every proper product-Gaussian posterior $\boldsymbol{H}$ and $t \in \{K+1, \dots, T\}$, the EBI of arm i is defined by

$$B_i(\boldsymbol{H}, t) := L(\boldsymbol{H}, t+1) - L_i(\boldsymbol{H}, t). \tag{9}$$

The value $L(\boldsymbol{H}(t-1), t+1)$ represents the loss function when a sample at round t is discarded.[5] It is a feasible policy in the definition of $L_i(\boldsymbol{H}, t)$ to ignore the observation from arm i and then run a policy attaining $L(\boldsymbol{H}, t+1)$. Consequently,

$$L_i(\boldsymbol{H}, t) \le L(\boldsymbol{H}, t+1), \qquad B_i(\boldsymbol{H}, t) \ge 0.$$

Therefore, the EBI indicates the decrease in the loss function due to a single sample from arm i. Since $L(\boldsymbol{H}(t-1), t+1)$ does not depend on arm i, maximizing of the action value function $-L_i(\boldsymbol{H}(t-1), t)$ is equivalent to maximizing $B_i(\boldsymbol{H}(t-1), t)$. We can now redefine the Bayes optimal algorithm in terms of the EBI:

Theorem 1. (Bayes optimal algorithm) For $t = K+1, \dots, T$, the flat-prior generalized Bayes optimal algorithm A^*_T is such that

$$I(t) \in \arg\max_{i \in [K]} B_i(\boldsymbol{H}(t-1), t)$$
$$J(T) \in \arg\max_{i \in [K]} \hat{\mu}_i(T),$$

and its conditional posterior expected simple regret at posterior $\boldsymbol{H}(t-1)$ at the beginning of round t is $L(\boldsymbol{H}(t-1), t)$. For every realized proper posterior $\boldsymbol{H}(K)$, this algorithm minimizes the conditional posterior simple regret in Eq. (3).

Proof. For a terminal posterior $\boldsymbol{H} = \bigotimes_i \mathcal{N}(m_i, 1/n_i)$, the posterior expected loss of recommending arm j is

$$\mathbb{E}_{\boldsymbol{H}}[\mu^*] - m_j.$$

It is therefore minimized exactly by the arms with maximal posterior mean, which proves the terminal assertion. Backward induction and Eq. (8) show that $L(\boldsymbol{H}, t)$ is the minimum posterior expected terminal

[5] Note that $t+1$ is indeed correctly placed in Eq. (9). It indicates one round fewer than t.

loss over all continuation policies with rounds $t, t+1, \ldots, T$ remaining, and that an action is Bayes optimal exactly when it minimizes $L_i(\boldsymbol{H}, t)$. Finally, Eq. (9) subtracts each $L_i(\boldsymbol{H}, t)$ from the same quantity $L(\boldsymbol{H}, t+1)$, so minimizing the former is equivalent to maximizing the EBI. Applying the recursion from the realized proper posterior $\boldsymbol{H}(K)$ proves Eq. (3). □

Thus, the Bayes optimal algorithm chooses an arm with the largest EBI.

2.6 One-step lookahead approximations

Since the Bayes optimal algorithm is difficult to calculate, several approximations have been considered. The knowledge-gradient rule is the one-step lookahead policy

$$I_{\mathrm{KG}}(t) \in \arg\max_{i\in[K]} \mathbb{E}_{X_i|\boldsymbol{H}(t-1)} \left[\max_{j\in[K]} \mathsf{m}_j\big(\mathsf{U}_i\big(\boldsymbol{H}(t-1), X_i\big)\big) - \max_{j\in[K]} \mathsf{m}_j\big(\boldsymbol{H}(t-1)\big) \right],$$

where, separately for each candidate arm i, the expectation is taken under the candidate-action posterior-predictive law in Eq. (4) [Gupta and Miescke, 1994, Frazier et al., 2008]. A corresponding posterior-mean expected-improvement rule retains only the positive improvement of the candidate arm over the current largest posterior mean:

$$I_{\mathrm{EI}}(t) \in \arg\max_{i\in[K]} \mathbb{E}_{X_i|\boldsymbol{H}(t-1)} \left[\left(\mathsf{m}_i\big(\mathsf{U}_i\big(\boldsymbol{H}(t-1), X_i\big)\big) - \max_{j\in[K]} \mathsf{m}_j\big(\boldsymbol{H}(t-1)\big) \right)_+ \right],$$

where $(x)_+ := \max\{x, 0\}$ [Jones et al., 1998, Bull, 2011].

Several known results demonstrate the suboptimality of KG and EI in K-armed BAI, which we describe in Section 6. In this paper, the convergence rate is characterized according to the Bayes optimal algorithm rather than its one-step approximations KG and EI. We show that the Bayes optimal algorithm performs very differently from the frequentist BAI, despite not making any myopic approximation.

3 Main Result: Suboptimality of the Bayes Optimal Algorithm

The following is this paper's main theorem.

Theorem 2. (Main theorem) Let $\Delta_G > 0$, let $K = 3$, and put $\boldsymbol{\mu}^{(0)} = (0, 0, \Delta_G)$. There are constants $c, C > 0$ and T_0, possibly depending on Δ_G but not on T or on the selected posterior-optimal policy, such that, for every odd $T \geq T_0$ and every flat-prior generalized Bayes optimal algorithm A_T^* of horizon T,

$$\mathrm{R}_{\boldsymbol{\mu}^{(0)}}(T, A_T^*) \geq cT^{-C}. \tag{10}$$

Consequently, for every horizon-indexed family $(A_T^*)_{T\geq 3}$ obtained by choosing one flat-prior generalized Bayes optimal algorithm at each horizon,

$$\liminf_{T\to\infty} -\frac{1}{T} \log \mathrm{R}_{\boldsymbol{\mu}^{(0)}}(T, A_T^*) = 0,$$

so no such family has a positive frequentist rate at $\boldsymbol{\mu}^{(0)}$.

The proof of Theorem 2 explicitly constructs an instance for which Eq. (10) holds. Notably, Theorem 2 implies the following.

Remark 2. Letting $\boldsymbol{\mu}^{(0)}$ be an instance satisfying Theorem 2, for a family of algorithms $(A_T^{\mathrm{freq}})_{T\geq 1}$ that has a positive frequentist rate at $\boldsymbol{\mu}^{(0)}$,

$$\limsup_{T\to\infty} \frac{\mathrm{R}_{\boldsymbol{\mu}^{(0)}}(T, A_T^*)}{\mathrm{R}_{\boldsymbol{\mu}^{(0)}}(T, A_T^{\mathrm{freq}})} = \infty,$$

with the ratio interpreted as $+\infty$ whenever its denominator is zero.

That is, the frequentist regret of the generalized Bayes optimal algorithm for this instance is arbitrarily worse than an algorithm with a frequentist rate (e.g., SR and uniform). Conditionally on every initialized posterior $\boldsymbol{H}(K)$, its posterior risk is no larger than that of any other continuation policy in $\mathcal{A}_{\mathrm{init}}$, by Definition 3.

Theorem 2 implies that the nature of the generalized Bayes algorithm A_T^* differs substantially from that of the algorithms designed for a frequentist measure. A frequentist guarantee seeks exponential convergence at every fixed unique-best instance, whereas Theorem 2 exhibits an instance at which A_T^* does not converge exponentially. The generalized Bayes algorithm A_T^* optimizes posterior expected performance at every initialized state and places greater weight on cases with a small gap between the optimal and suboptimal arms; it does not explore arms that only improve the higher-order polynomial of T^{-1}.

4 Proof of Theorem 2

The high-level idea behind the proof of Theorem 2 can be understood by considering a three-armed instance in which $\mu_1 = \mu_2 < \mu_3$. With a polynomially small probability in T^{-1}, arm 3 (the best arm) is underestimated, and consequently, $\hat{\mu}_1, \hat{\mu}_2 > \hat{\mu}_3$. When $\hat{\mu}_3$ is very small, the Bayes optimal algorithm allocates all samples to arms 1 and 2 because this more effectively reduce the Bayesian simple regret, which leads to underestimation of the best arm.

Formally, let $\Delta_G > 0$ be an arbitrary constant. Let $\boldsymbol{\mu}^{(0)} = (\mu_1^{(0)}, \mu_2^{(0)}, \mu_3^{(0)}) = (0, 0, \Delta_G)$. For ease of discussion, let T be odd and let T' be such that $2T' + 1 = T$. Furthermore, let $C_U > 0$ be such that

$$180 < \frac{(C_U - 6)^2}{4} \tag{11}$$

holds.[6] For each arm i and $n \geq 1$, define

$$S_{i,n} := \sum_{m=1}^{n} X_{i,m}, \qquad \hat{\mu}_{i,n} := \frac{S_{i,n}}{n}.$$

Whenever $N_i(t) = n$, the observed sufficient statistics satisfy $S_i(t) = S_{i,n}$ and $\hat{\mu}_i(t) = \hat{\mu}_{i,n}$. The variables with $n > N_i(t)$ are still well-defined by the reward stack, even though they have not been observed by

[6] It suffices that $C_U \geq 33$.

the algorithm. We also define the following events.

$$\begin{aligned}
\mathcal{X} &:= \{\hat{\mu}_{3,1} \le -C_U\sqrt{\log T}\},\\
\mathcal{Y}_i &:= \left\{|\hat{\mu}_{i,T'}| \le \frac{1}{T^2}\right\},\\
\mathcal{Z}_i &:= \bigcap_{n\in[T']} \left\{|\hat{\mu}_{i,n} - \hat{\mu}_{i,T'}| \le 3\sqrt{\log T}\sqrt{\sum_{m=n}^{T'-1}\frac{1}{m^2}}\right\},\\
\mathcal{E}_T &:= \mathcal{X}\cap\mathcal{Y}_1\cap\mathcal{Y}_2\cap\mathcal{Z}_1\cap\mathcal{Z}_2,\\
\mathcal{W}(t) &:= \mathcal{E}_T\cap\{N_1(t-1), N_2(t-1) \le T', N_3(t-1)\le 1\}.
\end{aligned}$$

Event $\mathcal{X}$ indicates that the first observation from the best arm, obtained during the fixed initialization, is a large underestimation. For $i = 1, 2$, event $\mathcal{Y}_i$ indicates that the empirical mean of arms i with T'-th draw is very close to the mean, and event $\mathcal{Z}_i$, $i = 1, 2$, indicates that the mean of arms i does not drift drastically. Event $\mathcal{W}(t)$ indicates that both arms 1 and 2 are not drawn in more than half the rounds and that arm 3 is only drawn once. For any two events $\mathcal{A}$ and $\mathcal{B}$, let $\mathcal{A},\mathcal{B} = \mathcal{A}\cap\mathcal{B}$.

Henceforth, probabilities of the events $\mathcal{X}$, $\mathcal{Y}_i$, $\mathcal{Z}_i$, and $\mathcal{W}(t)$ are under the frequentist model with true parameter $\boldsymbol{\mu}^{(0)}$. Probabilities explicitly associated with the recursive posterior evolution are instead under the indicated posterior-predictive law. Derivations of all lemmas appear in the appendices.

Lemma 3. (Underestimation of the best arm) For all sufficiently large T, we have

$$\mathbb{P}[\mathcal{X}] \ge \frac{T^{-\frac{(C_U+\Delta_G)^2}{2}}}{2(C_U+\Delta_G)\sqrt{2\pi\log T}} =: f_1.$$

Proof. Under $\boldsymbol{\mu}^{(0)}$, $\hat{\mu}_{3,1} \sim \mathcal{N}(\Delta_G, 1)$. Put $z_T = C_U\sqrt{\log T} + \Delta_G$. For all sufficiently large T, $z_T \ge \sqrt{2}$, and Lemma 10 gives

$$\mathbb{P}[\mathcal{X}] = \Phi^c(z_T) \ge \frac{1}{2z_T\sqrt{2\pi}}e^{-z_T^2/2}.$$

For $\log T \ge 1$,

$$z_T \le (C_U+\Delta_G)\sqrt{\log T}, \qquad z_T^2 \le (C_U+\Delta_G)^2\log T.$$

Substitution yields the claimed bound. □

Lemma 4. (Probability that the two means are close) For all sufficiently large T, the following holds:

$$\mathbb{P}[\mathcal{Y}_i] \ge \frac{1}{2}\sqrt{\frac{1}{\pi T^3}} =: f_2.$$

By using the fact that $\hat{\mu}_{i,T'} \sim \mathcal{N}(0, (T')^{-1})$, Lemma 4 follows from the tail bound of the normal distribution. Proofs of Lemma 4 and subsequent lemmas are found in the appendix.

Lemma 5. (Non-drastic drift) The following holds for each arm $i = 1, 2$:

$$\mathbb{P}[\mathcal{Y}_i, \mathcal{Z}_i^c] \le 2T^{-7/2} =: f_3.$$

We use $\mathrm{poly}(T^{-1})$ to describe a quantity that is polynomial in T^{-1}. According to Lemmas 3–5, $\mathcal{X}$, $\mathcal{Y}_1$, $\mathcal{Y}_2$, $\mathcal{Z}_1$, and $\mathcal{Z}_2$ occur with probability $\mathrm{poly}(T^{-1})$. The following lemmas bound the EBI under events $\mathcal{X}$, $\mathcal{Y}_1$, $\mathcal{Y}_2$, $\mathcal{Z}_1$, and $\mathcal{Z}_2$. All asymptotic bounds in Lemmas 6–8 and their supporting lemmas are uniform over the round t and over every reachable integer-precision posterior $\boldsymbol{H}(t-1)$ whose underlying reward-stack realization satisfies the stated event $\mathcal{W}(t)$; their constants may depend on C_U but not on t, T, or the realized posterior.

Lemma 6. Assume that $\mathcal{W}(t)$ holds. Then,

$$B_3(\boldsymbol{H}^{\mathrm{r}}(t-1),t) = O\left(T^{-\frac{(C_U-6)^2}{4}}\right) = o\left(T^{-180}\right).$$

Lemma 7. Assume that $\mathcal{W}(t)$ holds for a $t < T$. Then,

$$B_1(\boldsymbol{H}^{\mathrm{r}}(t-1),t), B_2(\boldsymbol{H}^{\mathrm{r}}(t-1),t) = \Omega(T^{-178}).$$

Moreover, if $N_j(t-1) = T'$, $j = 1, 2$, then[7]

$$B_j(\boldsymbol{H}^{\mathrm{r}}(t-1),t) < \max(B_1(\boldsymbol{H}^{\mathrm{r}}(t-1),t), B_2(\boldsymbol{H}^{\mathrm{r}}(t-1),t)).$$

Lemma 8. (Last round) Assume that $\mathcal{W}(T)$ holds. Then

$$B_1(\boldsymbol{H}^{\mathrm{r}}(T-1),T),\ B_2(\boldsymbol{H}^{\mathrm{r}}(T-1),T) = \Omega(T^{-22}),$$

and consequently $I(T) \neq 3$ for every generalized Bayes optimal action.

Lemma 6 provides an upper bound for $B_3(t)$ under $\mathcal{W}(t)$. Lemma 7 provides lower bounds for $B_1(t)$ and $B_2(t)$ under $\mathcal{W}(t)$.

Given these lemmas, we can provide the main theorem.

Proof of Theorem 2. By Theorem 1 and the null-set convention following Definition 3, the selected action is EBI-maximizing and the terminal recommendation is posterior-mean maximizing under the frequentist law $\mathbb{P}_{\boldsymbol{\mu}^{(0)}}$ almost surely. Accordingly, all event inclusions below are understood modulo $\mathbb{P}_{\boldsymbol{\mu}^{(0)}}$-null sets.

The event $\mathcal{E}_T$ implies $\mathcal{W}(4)$. We first show that, for $t \in \{4, 5, \ldots, T-1\}$, $\mathcal{W}(t)$ implies $\mathcal{W}(t+1)$. Lemmas 6 and 7 imply that $B_3(\boldsymbol{H}(t-1),t) < B_1(\boldsymbol{H}(t-1),t)$, so every EBI-maximizing action belongs to $\{1, 2\}$. Moreover, Lemma 7 shows that arm $i \in \{1, 2\}$ is not EBI-maximizing if $N_i(t-1) = T'$. Hence $\mathcal{W}(t+1)$ holds almost surely on $\mathcal{W}(t)$.

It follows that $\mathcal{W}(T)$ holds almost surely on $\mathcal{E}_T$. Lemmas 6 and 8 then imply that $I(T) \neq 3$ almost surely on $\mathcal{W}(T)$. At the beginning of round T, the two counts in $\{N_1(T-1), N_2(T-1)\}$ are T' and $T'-1$. Let $k \in \{1, 2\}$ be an arm that is not drawn at round T. Then $N_k(T) \in \{T'-1, T'\}$ and

$$\hat{\mu}_k(T) \geq -3\sqrt{\log T}\sqrt{\sum_{m=1}^{\infty}\frac{1}{m^2} - \frac{1}{T^2}} \quad (\text{by } \mathcal{Y}_k, \mathcal{Z}_k)$$

[7] Because $t < T$ and $N_3(t-1) = 1$, this implies $N_k(t-1) < T'$ for $k \in \{1, 2\} \setminus \{j\}$.

$$
\begin{aligned}
&\geq -\pi\sqrt{\frac{3\log T}{2}} - \frac{1}{T^2} \geq -6\sqrt{\log T} \\
&> -C_U\sqrt{\log T} \\
&\quad \text{(by Eq. (11))} \\
&\geq \hat{\mu}_{3,1} = \hat{\mu}_3(T).
\end{aligned}
$$

Since the terminal recommendation is posterior-mean maximizing almost surely, the strict inequality $\hat{\mu}_k(T) > \hat{\mu}_3(T)$ implies $J(T) \neq 3$ almost surely on $\mathcal{W}(T)$. Consequently,

$$
\mathbb{P}_{\boldsymbol{\mu}^{(0)}}(\mathcal{E}_T \cap \{J(T) = 3\}) = 0,
$$

and therefore $\mathbb{P}_{\boldsymbol{\mu}^{(0)}}[J(T) \neq 3] \geq \mathbb{P}_{\boldsymbol{\mu}^{(0)}}[\mathcal{E}_T]$. It follows that

$$
\begin{aligned}
\mathrm{R}_{\boldsymbol{\mu}^{(0)}}(T, A_T^*) &= \Delta_G \mathbb{P}_{\boldsymbol{\mu}^{(0)}}[J(T) \neq 3] \\
&\geq \mathbb{P}_{\boldsymbol{\mu}^{(0)}}[\mathcal{E}_T]\Delta_G \\
&\geq f_1(f_2 - f_3)^2\Delta_G, \qquad \text{(by Lemmas 3–5)}
\end{aligned}
$$

where the last step also uses independence of the reward sequences of the three arms. For all sufficiently large T,

$$
f_2 - f_3 \geq \frac{1}{4\sqrt{\pi}} T^{-3/2}, \qquad \frac{1}{\sqrt{\log T}} \geq T^{-1}.
$$

Substituting these two inequalities into the preceding probability lower bound shows that, for a constant $c > 0$ independent of T, the right-hand side is at least cT^{-C} with, for example, $C = (C_U + \Delta_G)^2/2 + 4$.

It remains to verify the stated consequence. Fix an arbitrary horizon-indexed family $(A_T^*)_{T \geq 3}$ satisfying the condition in the theorem, and put $r_T = \mathrm{R}_{\boldsymbol{\mu}^{(0)}}(T, A_T^*)$ and use the convention $-\log 0 = +\infty$. Since $0 \leq r_T \leq \Delta_G$ for every T,

$$
-\frac{1}{T}\log r_T \geq -\frac{1}{T}\log \Delta_G.
$$

On the odd subsequence, the polynomial lower bound gives

$$
-\frac{1}{T}\log r_T \leq \frac{C\log T - \log c}{T} \longrightarrow 0.
$$

The first display gives a global asymptotic lower bound of zero, and the odd subsequence gives the matching upper bound for the limit inferior. Hence the limit inferior in the theorem is zero. □

5 Proof Sketches of Lemmas 6 and 7

(Note: This section is moved from Appendix to Main Paper)

This section describes the main ideas of the proofs of Lemmas 6 and 7. Lemmas 6 and 7 are very different from the previous lemmas. Whereas Lemmas 3–5 deal with probabilities under the true parameter $\boldsymbol{\mu}^{(0)}$, Lemmas 6 and 7 are about how the Bayes optimal algorithm values the sample from each arm i at round t, which is represented by the recursive formula of Eq. (7). In particular, the latter lemmas

bound the EBI $B_i(\boldsymbol{H}(t-1),t) = L(\boldsymbol{H}(t-1),t+1) - L_i(\boldsymbol{H}(t-1),t)$ given the true posterior $\boldsymbol{H}(t-1)$. We first describe the evolution of $\boldsymbol{H}^{\mathrm{r}}(t-1)$ in Section 5.1. Derivation of Lemmas 6 and 7 are sketched in Sections 5.2 and 5.3, respectively.

5.1 Evolution of the posterior in the loss function

This section describes the evolution of $\boldsymbol{H}^{\mathrm{r}}(s)$ for $s \geq t-1$. At the beginning of round t, the Bayes optimal algorithm only observes $I(1), X(1), I(2), X(2), \ldots, I(t-1), X(t-1)$, which were drawn from the true distribution:[8]

$$X(t') \sim \mathcal{N}(\mu_{I(t')}, 1).$$

The value $L_i(\boldsymbol{H}(t-1),t)$ involves future rewards $I^{\mathrm{r}}(t), X^{\mathrm{r}}(t), I^{\mathrm{r}}(t+1), X^{\mathrm{r}}(t+1), \ldots, I^{\mathrm{r}}(T), X^{\mathrm{r}}(T)$, which are random variables that are generated under the posterior-predictive law induced by $\boldsymbol{H}(t-1)$, rather than under the fixed true parameter.[9] Namely, for $s \geq t-1$ and $i = I(s+1)$,

$$\begin{aligned}
\tilde{\mu}_i(s) &\sim \mathcal{N}\left(\hat{\mu}_i^{\mathrm{r}}(s), \frac{1}{N_i^{\mathrm{r}}(s)}\right) \\
X^{\mathrm{r}}(s+1) &\sim \mathcal{N}(\tilde{\mu}_i(s), 1) \\
\hat{\mu}_i(s+1) &= \frac{N_i^{\mathrm{r}}(s)\hat{\mu}_i^{\mathrm{r}}(s) + X^{\mathrm{r}}(s+1)}{N_i^{\mathrm{r}}(s)+1},
\end{aligned}$$

or equivalently,

$$\hat{\mu}_i^{\mathrm{r}}(s+1) - \hat{\mu}_i^{\mathrm{r}}(s) \sim \mathcal{N}\left(0, \frac{1}{N_i^{\mathrm{r}}(s)(N_i^{\mathrm{r}}(s)+1)}\right)$$

and the posterior is updated accordingly. In other words, drawing arm i with its posterior $\mathcal{N}(\hat{\mu}_i^{\mathrm{r}}(s), 1/N_i^{\mathrm{r}}(s))$ reduces its variance from $1/N_i^{\mathrm{r}}(s)$ to $1/(N_i^{\mathrm{r}}(s)+1)$.

5.2 Sketch of the proof of Lemma 6

Lemma 6 provides an upper bound for the EBI of arm 3 when the posterior probability of the arm being best is small. In the proof of Lemma 6, we derive Lemmas 11 and 12. Lemma 11 bounds the exclusion loss

$$\mathrm{R}^{(3)}_{\boldsymbol{H}^{\mathrm{r}}(t-1)} = \mathbb{E}_{\boldsymbol{H}^{\mathrm{r}}(t-1)}\left[\left(\mu_3 - \min_{j\neq 3}\mu_j\right)_+\right].$$

This exclusion loss also upper-bounds

$$\mathbb{E}_{\boldsymbol{H}^{\mathrm{r}}(t-1)}[\mathbf{1}[i^*(\boldsymbol{\mu}) = 3]\mathrm{R}_{\boldsymbol{\mu}}(T)],$$

the portion of the Bayesian simple regret arising when arm 3 is optimal. Lemma 12 then uses the exclusion loss to bound the EBI.

The greatest challenge in bounding the EBI $B_i(\boldsymbol{H}^{\mathrm{r}}(t-1),t) = L(\boldsymbol{H}^{\mathrm{r}}(t-1),t+1) - L_i(\boldsymbol{H}^{\mathrm{r}}(t-1),t)$ lies in its adaptivity. A slight difference in the posterior $\boldsymbol{H}^{\mathrm{r}}(s)$ can change the selection sequence

[8] Note that we consider the frequentist setting in which the (unknown) true parameters exist.

[9] The Bayes optimal algorithm does not know the true parameter $\boldsymbol{\mu}$ and instead evaluates future outcomes using the posterior $\boldsymbol{H}(t-1)$.

$I^{\mathrm{r}}(s), I^{\mathrm{r}}(s+1), \dots, I^{\mathrm{r}}(T)$, and thus it is generally difficult to compute the EBI exactly. The high-level idea of Lemma 12 is as follows. When an arm is very unlikely to be the best arm (i.e., under event $\mathcal{X}$), the next sample for that arm does not significantly improve the simple regret. Notice that the difference between $L(\boldsymbol{H}^{\mathrm{r}}(t-1), t+1)$ and $L_i(\boldsymbol{H}^{\mathrm{r}}(t-1), t)$ is derived from the draw for arm i at round t. To bound the EBI, we introduce a *mock-sample* algorithm, along with its loss $L^{\mathrm{mock}}(\boldsymbol{H}^{\mathrm{r}}(t-1), t)$, which utilizes the external randomizer $U^{\mathrm{r}}(t)$ such that

$$\begin{aligned}
&U^{\mathrm{r}}(t), I^{\mathrm{r}}(t+1), X^{\mathrm{r}}(t+1), I^{\mathrm{r}}(t+2), X^{\mathrm{r}}(t+2), \dots, I^{\mathrm{r}}(T), X^{\mathrm{r}}(T) \\
&\qquad \text{of } L^{\mathrm{mock}}(\boldsymbol{H}^{\mathrm{r}}(t-1), t), \text{ and} \\
&X^{\mathrm{r}}(t), I^{\mathrm{r}}(t+1), X^{\mathrm{r}}(t+1), I^{\mathrm{r}}(t+2), X^{\mathrm{r}}(t+2), \dots, I^{\mathrm{r}}(T), X^{\mathrm{r}}(T) \\
&\qquad \text{of } L_i(\boldsymbol{H}^{\mathrm{r}}(t-1), t)
\end{aligned}$$

have exactly the same distribution. Setting $i = 3$, we have

$$\begin{aligned}
B_3(\boldsymbol{H}^{\mathrm{r}}(t-1), t) &:= L(\boldsymbol{H}^{\mathrm{r}}(t-1), t+1) - L_3(\boldsymbol{H}^{\mathrm{r}}(t-1), t) \\
&\le L^{\mathrm{mock}}(\boldsymbol{H}^{\mathrm{r}}(t-1), t) - L_3(\boldsymbol{H}^{\mathrm{r}}(t-1), t) \\
&\quad \text{(by optimality of } L(\boldsymbol{H}^{\mathrm{r}}(t-1), t+1)) \\
&\le \mathrm{R}^{(3)}_{\boldsymbol{H}^{\mathrm{r}}(t-1)}.
\end{aligned}$$

5.3 Sketch of the proof of Lemma 7

Lemma 7 provides a lower bound for the EBI $B_i(\boldsymbol{H}^{\mathrm{r}}(t-1), t) = L(\boldsymbol{H}^{\mathrm{r}}(t-1), t+1) - L_i(\boldsymbol{H}(t-1), t)$ for $i = 1, 2$. Note that we define the events $\mathcal{Y}_1$, $\mathcal{Y}_2$, $\mathcal{Z}_1$, and $\mathcal{Z}_2$ such that $\hat{\Delta}(t) := \hat{\mu}_1(t) - \hat{\mu}_2(t)$ is small. If $\hat{\mu}_1(t) - \hat{\mu}_2(t)$ is small enough and arm 3 is sufficiently worse than the other two arms, we can expect that drawing arm 1 or 2 reduces the loss function by $\mathrm{poly}(T^{-1})$. Again, evaluating $L(\boldsymbol{H}^{\mathrm{r}}(t-1), t+1) - L_i(\boldsymbol{H}^{\mathrm{r}}(t-1), t)$ exactly is very challenging in general, as the draws $I^{\mathrm{r}}(t), I^{\mathrm{r}}(t+1), I^{\mathrm{r}}(t+2), \dots, I^{\mathrm{r}}(T)$ depend on the recursive computation of the future rewards. We solve this problem in steps as follows:

(A) First, when the probability of arm 3 being best is very small (i.e., under event $\mathcal{X}$), we can show that

$$|L(\boldsymbol{H}^{\mathrm{r}}(t-1), t+1) - L(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t+1)|$$

is also small, where $\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1)$ is the posterior in the two-armed instance with arm 3 removed. In other words, we have reduced a three-armed instance to a two-armed instance (Section E.1).

(B) Second, in the two-armed instance, there exists a Bayes optimal continuation that always draws a smaller-count arm, with ties resolved by a fixed count-only rule. Therefore, we can explicitly obtain the distribution of $\hat{\Delta}^{\mathrm{r}}(T-1)$ given $\hat{\Delta}(t-1)$. In the two-armed problem, we show that $\mathcal{W}(t)$ implies a sufficiently small $\hat{\Delta}(t-1)$, which yields a sufficiently small $\hat{\Delta}^{\mathrm{r}}(T-1)$ with probability $\Omega(\mathrm{poly}(T^{-1}))$, providing an $\Omega(\mathrm{poly}(T^{-1}))$ value for $L(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1), T+1) - L_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1), T)$. These results yield an $\Omega(\mathrm{poly}(T^{-1}))$ lower bound for the EBI $B_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t)$ in the two-armed instance (Section E.2).

(C) Finally, by using (A) and (B), we derive Lemma 7 (Section E.3).

As a byproduct of the analysis, we have obtained an explicit count-balancing Bayes optimal continuation for the two-armed problem.

Remark 3. (Characterization of the Bayes optimal algorithm for $K = 2$) In the reduced two-armed state considered here, Lemma 17 implies that $L_1(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t) = L_2(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t)$ if $N_1(t-1), N_2(t-1) < T'$. If $N_1(t-1) = T'$ and $N_2(t-1) < T'$, then $L_1(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t) > L_2(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t)$. Consequently, with the remaining budget in this reduced problem, a Bayes optimal continuation balances the terminal counts at T' each.

6 Related Work

This section reviews related studies concerning Bayesian algorithms in the context of sequential decision-making. The multi-armed bandit problem [Robbins, 1952] involves multiple treatment arms and decisions made using samples obtained sequentially via experiments. The aim is to maximize the sum of the rewards, which boils down to cumulative regret minimization (CRM). Although this problem has attracted considerable attention from the machine learning community, this study has focused on another established branch of bandit problems, BAI, in which the goal is to identify the treatment arm with the largest mean. Finding the best treatment essentially concerns simple regret minimization (SRM). The different goals of these two approaches results in significant differences between the approaches of CRM and SRM algorithms to balancing exploration and exploitation.

Best arm identification (SRM): Although the term "best arm identification" was coined in the early 2010s [Audibert et al., 2010, Bubeck et al., 2011], similar ideas had already attracted substantial attention in various fields [Paulson, 1964, Maron and Moore, 1997, Even-Dar et al., 2006]. Audibert et al. [2010] proposed the successive rejects algorithm, which demonstrates frequentist simple regret with the order of $\exp(-\Gamma(\boldsymbol{\mu})T)$ for some $\Gamma(\boldsymbol{\mu})$. Elsewhere, Carpentier and Locatelli [2016] showed an instance $\boldsymbol{\mu}$ for which this $\Gamma(\boldsymbol{\mu})$ is optimal up to a constant factor. For a fixed instance with a unique best arm and a positive minimum gap, the probability of error $\mathbb{P}[J(T) \neq i^*]$ has the same exponential rate as the frequentist simple regret (Audibert et al. 2010, Section 2), and is also commonly used to evaluate a BAI algorithm.

The aforementioned works focus on fixed-budget identification, in which the horizon T is fixed. There is also a significant body of literature devoted to fixed-confidence identification. In this scenario, the forecaster is given a confidence level δ and aims to stop the sampling as soon as they are confident in the identification with a probability of $1-\delta$. For fixed-confidence identification, several algorithms are known, such as successive elimination [Even-Dar et al., 2006], track and stop [Garivier and Kaufmann, 2016], and top-two algorithms [Russo, 2020, Shang et al., 2020, Jourdan et al., 2022, You et al., 2023]. The designs of fixed-confidence and fixed-budget algorithms are markedly different [Carpentier and Locatelli, 2016, Komiyama et al., 2022, Degenne, 2023].

Ranking and selection: A particularly interesting strand of research concerns the RS problem [Hong et al., 2021]. RS and BAI have substantial overlap: RS includes static, multistage, and fully sequential procedures, and its fixed-budget formulation can itself be expressed as a dynamic program. One influential RS approximation is the optimal computing budget allocation approach initiated by Chen et al. [2000].

In its oracle static form, an allocation is chosen to maximize the probability of correct selection:

$$\max_{\substack{N_j \in \mathbb{N},\, N_j \geq 1 \\ \sum_j N_j = T}} \mathbb{P}_{\boldsymbol{\mu}} \left(\hat{\mu}_{i^*}(N_{i^*}) > \max_{j \neq i^*} \hat{\mu}_j(N_j) \right). \tag{12}$$

The oracle objective depends on the unknown model parameters; practical OCBA procedures replace them with estimates and update the allocation as data are collected. Elsewhere, Glynn and Juneja [2004] observed an asymptotically optimal allocation for Eq. (12) based on the large deviation theory. Additional adaptive algorithms have been developed based on Bayesian ideas. For example, the seminal work by Chick and Inoue [2001] proposed a two-stage method called the expected value of information algorithm, in which the information from the first stage is used as a prior to optimize the allocation of the second stage. The KG algorithm [Gupta and Miescke, 1994] also adopts a Bayesian approach. However, the analytical properies of KG are little understood. The results of Ryzhov et al. [2012] show the discounted version of KG most frequently drawing the best arm. Elsewhere, Wang and Powell [2018] have further characterized KG without giving a rate for the simple regret. Another well-known one-step lookahead algorithm is EI [Jones et al., 1998, Bull, 2011]. Ryzhov [2016] showed that EI draws suboptimal arms $\Theta(\log T)$ times, implying the suboptimality of EI in terms of the frequentist simple regret (or probability of error). Qin et al. [2017] and Chen and Ryzhov [2019] proposed modified versions of EI with an exponential rate of convergence.

Bayesian algorithms for CRM: Thompson sampling [Thompson, 1933] is among the oldest of the heuristics and is known to be asymptotically optimal in terms of the frequentist CRM [Agrawal and Goyal, 2012, Kaufmann et al., 2012, Komiyama et al., 2015]. One of the seminal results regarding Bayesian CRM is the Gittins index theorem [Gittins, 1989, Weber, 1992], which states that minimizing the discounted Bayesian regret can be achieved by computing the Gittins indices of individual arms.[10] The Bayes optimal algorithms for CRM and SRM differ considerably. In the former setting, the algorithm obtains a reward at the end of each round. In the latter setting, the corresponding reward is delayed until round T, which makes our problem particularly challenging.

7 Discussion

This paper has considered the BAI problem. We have demonstrated that the flat-prior generalized Bayes algorithm, which is pointwise optimal for posterior expected performance after the fixed initialization, does not have a frequentist rate of simple regret. For some frequentist instances, posterior optimization assigns too little value to recovering an arm that was severely underestimated during initialization. A high-level implication is that if an approximate Bayesian algorithm exhibits a uniform exponential convergence, this results not from considering a lookahead but from some idea that lends it the robustness preferred by frequentists. A challenge in analyzing a sequential decision-making algorithm is its flexibility. The decision $I(t)$ at round t depends on the results of the Bellman equation, which is difficult to compute exactly. Accordingly, we have introduced a quantity called the EBI, which represents the improvement of the Bayesian simple regret from a single sample. We have introduced several novel ideas for analyzing

[10] Note that the Gittins index is no longer optimal in the context of undiscounted regret [Berry and Fristedt, 1985]; see also Section 1.3.3 in Kaufmann [2014].

the EBI and obtained an explicit count-balancing Bayes optimal continuation for two-armed BAI as a byproduct.

It is worth remarking on the unusual instance from which we derived Theorem 2. We have provided an instance in which $\mu_1 = \mu_2$. The two suboptimal arms are then statistically tied, so there is no uniquely better one to distinguish; posterior uncertainty about their ordering nevertheless keeps the marginal value of sampling both arms significant. Our proof exploits this exact tie and does not establish the same phenomenon when all means are distinct. This is a limitation of our results and highlights the following open problem.

Conjecture 9. (Open problem: Exponential convergence of the Bayes optimal algorithm for almost all parameters) Under some regularity conditions over the prior, the Bayes optimal algorithm has a regret that falls exponentially with T for almost all the instances.

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A General Lemma

Lemma 10 (Normal tail bound; Feller 1968)**.** Let $\phi(x) := e^{-x^2/2}/\sqrt{2\pi}$ be the probability density function (pdf) of a standard normal random variable. Let $\Phi^c(x) = \int_x^\infty \phi(x')dx'$. Then, for any $x > 0$, the following inequality holds:

$$\left(\frac{1}{x} - \frac{1}{x^3}\right)\frac{e^{-x^2/2}}{\sqrt{2\pi}} \leq \Phi^c(x) \leq \frac{1}{x}\frac{e^{-x^2/2}}{\sqrt{2\pi}}.$$

B Proof of Lemma 4

Since $\hat{\mu}_{i,T'}$ is a zero-mean normal random variable with variance ${T'}^{-1}$ and $T' = (T-1)/2 \geq T/3$ for $T \geq 3$, we have

$$\begin{aligned}\mathbb{P}\left[|\hat{\mu}_{i,T'}| \leq \frac{1}{T^2}\right] &= 2\sqrt{\frac{T'}{2\pi}}\int_0^{T^{-2}} \exp(-T'x^2/2)\,dx \\ &\geq \frac{2}{T^2}\sqrt{\frac{T}{6\pi}}\exp\left(-\frac{T'}{2T^4}\right) \geq \frac{1}{2}\sqrt{\frac{1}{\pi T^3}}\end{aligned}$$

for all sufficiently large T.

C Proof of Lemma 5

For $n \leq T'$, put

$$D_n := \hat{\mu}_{i,n} - \hat{\mu}_{i,T'}, \qquad v_n := \frac{1}{n} - \frac{1}{T'}.$$

Then $D_n \sim \mathcal{N}(0, v_n)$, and

$$\sum_{m=n}^{T'-1}\frac{1}{m^2} \geq \sum_{m=n}^{T'-1}\frac{1}{m(m+1)} = v_n.$$

Consequently, a Gaussian tail bound and a union bound give

$$\begin{aligned}\mathbb{P}[\mathcal{Y}_i, \mathcal{Z}_i^c] &\leq \sum_{n=1}^{T'-1}\mathbb{P}\left[|D_n| > 3\sqrt{\log T}\sqrt{\sum_{m=n}^{T'-1} m^{-2}}\right] \\ &\leq 2T'\exp\left(-\frac{9}{2}\log T\right) \leq 2T^{-7/2}.\end{aligned}$$

This proves Lemma 5.

D Proof of Lemma 6

In this section, we obtain an upper bound for $B_i = L(\boldsymbol{H}^{\mathrm{r}}(t-1), t+1) - L_i(\boldsymbol{H}^{\mathrm{r}}(t-1), t)$. The gap between $L(\boldsymbol{H}^{\mathrm{r}}(t-1), t+1)$ and $L_i(\boldsymbol{H}^{\mathrm{r}}(t-1), t)$ is derived from the draw of arm i at round t. However, bounding the difference is challenging because a slight change in the posterior can drastically affect the subsequent choice of arms. We introduce a mock-sample loss that simulates the Bayes optimal algorithm with a slightly modified posterior so that the difference between $\boldsymbol{H}^{\mathrm{r}}(t-1)$ and $\boldsymbol{H}^{\mathrm{r}}(t)$ is absorbed. In the proof, we derive Lemmas 11 and 12. Lemma 6 is a straightforward corollary of these two lemmas.

Lemma 11. For a posterior $\boldsymbol{H}$ and an arm i, define the exclusion loss

$$\mathrm{R}^{(i)}_{\boldsymbol{H}} := \mathbb{E}_{\boldsymbol{H}}\left[\left(\mu_i - \min_{j\neq i}\mu_j\right)_+\right].$$

Assume that event $\mathcal{W}(t)$ holds. Then,

$$\mathrm{R}^{(3)}_{\boldsymbol{H}^{\mathrm{r}}(t-1)} \leq g_0$$

where $g_0 = g_0(T, C_U)$ is $O\left(T^{-\frac{(C_U-6)^2}{4}}\right)$.

Proof of Lemma 11. We first bound $\hat{\mu}_1(t)$ and $\hat{\mu}_2(t)$ from below:

$$\begin{aligned}
\hat{\mu}_i(t-1) &= \hat{\mu}_{i,N_i(t-1)} \\
&\geq -3\sqrt{\log T}\sqrt{\sum_{m=1}^{\infty}\frac{1}{m^2}} - \frac{1}{T^2} \\
&\quad (\text{by } \mathcal{Y}_i\mathcal{Z}_i) \\
&\geq -\pi\sqrt{\frac{3\log T}{2}} - \frac{1}{T^2} \\
&\geq -6\sqrt{\log T}.
\end{aligned}$$

Let $C_F = 6$, $x \sim \mathcal{N}(-C_U\sqrt{\log T}, 1)$, and $y \sim \mathcal{N}(-C_F\sqrt{\log T}, 1)$ be independent. Under $\mathcal{W}(t)$, the posterior mean of μ_3 is at most that of x, and the posterior mean of each μ_i, $i = 1, 2$, is at least that of y. The fixed initialization gives $N_k(t-1) \geq 1$ for every arm (and $N_3(t-1) = 1$ under $\mathcal{W}(t)$), so all posterior variances are at most one. Adding independent zero-mean Gaussian noise if necessary and using that $z \mapsto z_+$ is increasing and convex, we obtain

$$\begin{aligned}
&\mathrm{R}^{(3)}_{\boldsymbol{H}^{\mathrm{r}}(t-1)} \\
&\leq \sum_{i=1}^{2}\mathbb{E}_{\boldsymbol{H}^{\mathrm{r}}(t-1)}[(\mu_3 - \mu_i)_+] \\
&\leq 2\mathbb{E}[(x-y)_+] \\
&= \frac{2}{\sqrt{4\pi}}\int_0^{\infty} z\exp\left(-\frac{(z+(C_U-C_F)\sqrt{\log T})^2}{4}\right)dz \\
&\leq \frac{2}{\sqrt{\pi}}T^{-(C_U-C_F)^2/4} =: g_0,
\end{aligned}$$

which is $O\left(T^{-\frac{(C_U-C_F)^2}{4}}\right) = O\left(T^{-\frac{(C_U-6)^2}{4}}\right)$. □

Lemma 12. (Upper bound of the EBI) For any $t \in \{K+1, \ldots, T\}$, the following inequality holds:

$$B_i(\boldsymbol{H}^{\mathrm{r}}(t-1), t) \leq \mathrm{R}^{(i)}_{\boldsymbol{H}^{\mathrm{r}}(t-1)}.$$

Proof of Lemma 12. By definition,

$$B_i(\boldsymbol{H}^{\mathrm{r}}(t-1), t) := L(\boldsymbol{H}^{\mathrm{r}}(t-1), t+1) - L_i(\boldsymbol{H}^{\mathrm{r}}(t-1), t).$$

We first introduce the notion of mock-sample loss.

Definition 5. (Mock-sample loss) Independently of $\boldsymbol{\mu} \sim \boldsymbol{H}^{\mathrm{r}}(t-1)$, let

$$\mu_i' \sim \mathcal{N}\left(\hat{\mu}_i^{\mathrm{r}}(t-1), \frac{1}{N_i^{\mathrm{r}}(t-1)}\right),$$

$$U_{i,0}, U_{i,1}, \ldots \mid \mu_i' \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_i', 1), \qquad U^{\mathrm{r}}(t) := U_{i,0}.$$

That is, $U^{\mathrm{r}}(t)$ is a simulated sample that is obtained from the external randomizer. Moreover, let

$$(S_j^{\mathrm{r}'}, N_j^{\mathrm{r}'}) = \begin{cases} (S_i^{\mathrm{r}}(t-1) + U^{\mathrm{r}}(t), N_i^{\mathrm{r}}(t-1) + 1) & (j = i) \\ (S_j^{\mathrm{r}}(t-1), N_j^{\mathrm{r}}(t-1)) & (j \neq i) \end{cases}$$

be the updated posterior. Let π_i^{mock} be the randomized recursive policy that applies a measurable Bayes-optimal continuation from this mock-updated state and thereby determines the complete future adaptive sequence $I^{\mathrm{r}}(t+1), \ldots, I^{\mathrm{r}}(T)$ as if $(S_j^{\mathrm{r}'}, N_j^{\mathrm{r}'})_{j\in[K]}$ were $\boldsymbol{H}^{\mathrm{r}}(t)$, uses the corresponding posterior-predictive pseudo-stream whenever that internal policy samples arm i, and recommends an arm other than i. At each actual sampling opportunity it draws the arm requested by the internal policy. If that arm is i, it discards the actual observation and supplies the next $U_{i,r}$ to the internal policy; otherwise it supplies the actual observation. Put

$$J_i^{\mathrm{mock}}(T) := \min \arg\max_{j\in[K]\setminus\{i\}} \hat{\mu}_j^{\mathrm{r}}(T),$$
$$L^{\mathrm{mock}}(\boldsymbol{H}^{\mathrm{r}}(t-1), t) := \mathbb{E}^{\pi_i^{\mathrm{mock}}}_{\boldsymbol{H}^{\mathrm{r}}(t-1)} \left[\mu^* - \mu_{J_i^{\mathrm{mock}}(T)} \right],$$

where the expectation includes the posterior-predictive continuation and all auxiliary randomization. Thus $\pi_i^{\mathrm{mock}} \in \Pi_{t+1}(\boldsymbol{H}^{\mathrm{r}}(t-1))$: its actual sampling opportunities are $t+1, \ldots, T$, whereas $U_{i,0}$ is auxiliary randomization rather than an actual sample.

Then, the following inequalities hold:

$$L(\boldsymbol{H}^{\mathrm{r}}(t-1), t+1) \leq L^{\mathrm{mock}}(\boldsymbol{H}^{\mathrm{r}}(t-1), t) \leq L_i(\boldsymbol{H}^{\mathrm{r}}(t-1), t) + \mathrm{R}^{(i)}_{\boldsymbol{H}^{\mathrm{r}}(t-1)}.$$

The first inequality follows because the mock-sample rule is a feasible randomized policy with only $T-t$ actual samples. Whenever its internal rule asks for a sample from arm i, the actual observation can be discarded and a posterior-predictive pseudo-observation can be generated by the external randomizer. Thus $L(\boldsymbol{H}^{\mathrm{r}}(t-1), t+1)$, which is the minimum risk among all such policies, cannot be larger.[11]

For the second inequality, couple this pseudo-stream with the real-sample policy defining $L_i(\boldsymbol{H}^{\mathrm{r}}(t-1), t)$, and modify only its terminal recommendation so that it never recommends arm i. The posterior is a product across arms. Consequently, the joint law of the arm-i stream and the parameters and observations of all other arms is unchanged when the real arm-i stream is replaced by its independent posterior-predictive copy. For every recommendation $J \neq i$, the loss admits the decomposition

$$\mu^* - \mu_J = \max_{j\neq i} \mu_j - \mu_J + \left(\mu_i - \max_{j\neq i} \mu_j \right)_+ .$$

The second term does not depend on the sampling or recommendation rule, while the joint distribution relevant to the first term is identical under the two coupled policies. Hence the mock loss equals the loss

[11] Equivalently, at time $t-1$ the external randomizer first generates an independent copy μ_i' from the posterior of μ_i and then generates the entire pseudo-reward stream of arm i conditionally on μ_i'. This stream has the same posterior-predictive law as the real reward stream of arm i.

of the real-sample policy with arm i excluded only from its terminal recommendation.

If the original policy recommends an arm other than i, the modification does not change its recommendation. If it recommends i, the increase in loss is at most $(\mu_i - \min_{j\neq i}\mu_j)_+$. Therefore,

$$\begin{aligned} L^{\mathrm{mock}}(\boldsymbol{H}^{\mathrm{r}}(t-1),t) - L_i(\boldsymbol{H}^{\mathrm{r}}(t-1),t) &\leq \mathbb{E}_{\boldsymbol{H}^{\mathrm{r}}(t-1)}\left[\left(\mu_i - \min_{j\neq i}\mu_j\right)_+\right] \\ &= \mathrm{R}^{(i)}_{\boldsymbol{H}^{\mathrm{r}}(t-1)}. \end{aligned}$$

□

Proof of Lemma 6. Lemma 6 directly follows from Lemmas 11 and 12 with $i = 3$. □

E Proof of Lemma 7

This section derives a lower bound for $B_i(\boldsymbol{H}^{\mathrm{r}}(t-1),t) := L(\boldsymbol{H}^{\mathrm{r}}(t-1),t+1) - L_i(\boldsymbol{H}^{\mathrm{r}}(t-1),t)$ for arms $i = 1, 2$.

E.1 Reduction to a two-armed problem

The following lemma bounds the gap between the three-armed best arm identification problem and the two-armed best arm identification problem in which one of the arms has been removed.

Lemma 13. (Reduction to a two-armed problem) Let $\boldsymbol{H}^{\mathrm{r}}_{\setminus i}$ be the posterior of the two-armed instance in which arm i has been removed. The following inequality holds:

$$|L(\boldsymbol{H}^{\mathrm{r}}(t-1),t+1) - L(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1),t+1)| \leq \mathrm{R}^{(3)}_{\boldsymbol{H}^{\mathrm{r}}(t-1)}. \tag{13}$$

Similarly, we have

$$|L_j(\boldsymbol{H}^{\mathrm{r}}(t-1),t) - L_j(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1),t)| \leq \mathrm{R}^{(3)}_{\boldsymbol{H}^{\mathrm{r}}(t-1)}. \tag{14}$$

for $j \in \{1,2\}$.

Proof of Lemma 13. Put $E_3 = \mathrm{R}^{(3)}_{\boldsymbol{H}^{\mathrm{r}}(t-1)}$. The following coupling argument applies with any fixed number of remaining samples. Embed an optimal two-armed policy into the three-armed problem by never sampling or recommending arm 3. If its recommendation is $J \in \{1,2\}$, then pointwise

$$\begin{aligned} \max_{k\in[3]}\mu_k - \mu_J &= \max_{k\in\{1,2\}}\mu_k - \mu_J + \left(\mu_3 - \max_{k\in\{1,2\}}\mu_k\right)_+ \\ &\leq \max_{k\in\{1,2\}}\mu_k - \mu_J + \left(\mu_3 - \min_{k\in\{1,2\}}\mu_k\right)_+. \end{aligned}$$

Taking posterior expectations and then the infimum over three-armed policies gives, at either stage $s = t$ or $s = t+1$,

$$L(\boldsymbol{H}^{\mathrm{r}}(t-1),s) - L(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1),s) \leq E_3.$$

Conversely, in the two-armed problem simulate an independent posterior-predictive stream for arm 3 and run an optimal three-armed policy internally. Whenever that policy requests arm 3, consume the

round by drawing either real arm and discard that real observation. If the internal terminal recommendation is in $\{1,2\}$, keep it; if it is arm 3, replace it by a posterior-best arm in $\{1,2\}$. Coupling the imaginary arm-3 parameter and stream with their genuine three-armed counterparts, the pointwise increase in loss is at most $(\mu_3 - \min_{k\in\{1,2\}} \mu_k)_+$. Hence

$$L(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), s) - L(\boldsymbol{H}^{\mathrm{r}}(t-1), s) \le E_3.$$

Combining the last two displays with $s = t+1$ proves Eq. (13). Repeating the same two embeddings while requiring the first action at stage t to be $j \in \{1,2\}$ proves Eq. (14). □

E.2 Bounds on the two-armed problem

This subsection considers the loss of the two-armed best arm identification problem in which arm 3 has been removed. The goal here is to show that $B_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t)$ is $\Omega(\mathrm{poly}(T^{-1}))$ if $\hat{\Delta}^{\mathrm{r}}(t-1) := \hat{\mu}^{\mathrm{r}}_1(t-1) - \hat{\mu}^{\mathrm{r}}_2(t-1)$ is sufficiently small. To achieve this, we take the following steps. First, Section E.2.1 considers $B_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1), T)$ at round T (i.e., the last round); we show that $B_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1), T)$ is $\Omega(\mathrm{poly}(T^{-1}))$ when $\hat{\Delta}^{\mathrm{r}}(T-1)$ is small. Second, Section E.2.2 shows that there exists a Bayes optimal continuation in the two-armed problem that draws a smaller-count arm at every round and resolves ties by a fixed count-only rule. This selected continuation has a deterministic allocation sequence, which enables us to evaluate the distribution of $\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1)$ given $\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1)$. Third, Section E.2.3 shows that $\mathcal{W}(t)$ implies that $\hat{\Delta}(t-1)$ is sufficiently small that $B_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t)$ is $\Omega(\mathrm{poly}(T^{-1}))$.

E.2.1 Marginal loss reduction

First, the following lemma evaluates the simple regret given $\hat{\Delta}^{\mathrm{r}}(T)$ (i.e., the simple regret after T draws).

Lemma 14. (Evaluation of the Bayesian simple regret after T rounds) Let r be a zero-mean normal random variable with variance $V_r = 1/(N^{\mathrm{r}}_1(T)) + 1/(N^{\mathrm{r}}_2(T))$. Then,

$$L(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T), T+1) := \mathbb{E}_{\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T)}[\Delta_{J^{\mathrm{r}}(T)}] = \frac{1}{\sqrt{2\pi V_r}} \int_{-\infty}^{\infty} (r - |\hat{\Delta}^{\mathrm{r}}(T)|)_+ \exp\left(-\frac{r^2}{2V_r}\right) dr.$$

Proof. In the posterior $\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T)$, μ_i, for $i = 1, 2$, are distributed as:

$$\mu_i \sim \mathcal{N}\left(\hat{\mu}^{\mathrm{r}}_i(T), \frac{1}{N^{\mathrm{r}}_i(T)}\right)$$

and thus for $\{i, j\} = \{1, 2\}$,

$$(\mu_i - \mu_j) - (\hat{\mu}^{\mathrm{r}}_i(T) - \hat{\mu}^{\mathrm{r}}_j(T)) \sim \mathcal{N}(0, V_r). \tag{15}$$

Eq. (15) implies that the quantity $V_r = 1/(N^{\mathrm{r}}_1(T)) + 1/(N^{\mathrm{r}}_2(T))$ is the variance of $\mu_1 - \mu_2$ in the posterior, which we call posterior variance. Let $j = \widehat{J}(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T))$ and let i be the other arm. Then $\hat{\mu}^{\mathrm{r}}_j(T) \ge \hat{\mu}^{\mathrm{r}}_i(T)$ by Eq. (5). Arm j is recommended, and the simple regret is non-zero only when[12] $\mu_i > \mu_j$; namely,

$$\Delta_{J^{\mathrm{r}}(T)} = (\mu_i - \mu_j)_+$$

[12] That is, the signs of $\hat{\mu}^{\mathrm{r}}_j(T) - \hat{\mu}^{\mathrm{r}}_i(T)$ and $\mu_j - \mu_i$ are different.

and thus

$$\mathbb{E}_{\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T)}[\Delta_{J^{\mathrm{r}}(T)}] = \frac{1}{\sqrt{2\pi V_r}} \int_{-\infty}^{\infty} (x - |\hat{\Delta}^{\mathrm{r}}(T)|)_+ \exp\left(-\frac{x^2}{2V_r}\right) dx \quad \text{(by Eq. (15))}.$$

□

The following inequality quantifies the expected decrease in the loss function due to the last sample $i = I^{\mathrm{r}}(T)$.

Lemma 15. (One-step improvement with the T-th draw) The following inequality holds:

$$\begin{aligned} &L(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1), T+1) - L_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1), T) \\ &\qquad \geq \mathbb{P}\left[\frac{3}{2}|\hat{\Delta}^{\mathrm{r}}(T-1)| \leq l \leq 2|\hat{\Delta}^{\mathrm{r}}(T-1)|\right] \times \mathbb{P}\left[r \geq |\hat{\Delta}^{\mathrm{r}}(T-1)|\right] \times |\hat{\Delta}^{\mathrm{r}}(T-1)| \\ &\qquad\qquad =: g_1(\hat{\Delta}^{\mathrm{r}}(T-1), N_1^{\mathrm{r}}(T-1), N_2^{\mathrm{r}}(T-1), i) \geq 0. \end{aligned}$$

where l and r are independent zero-mean normal random variables with respective variances

$$\begin{aligned} V_l &= \frac{1}{N_i^{\mathrm{r}}(T-1)(1 + N_i^{\mathrm{r}}(T-1))}, \\ V_r &= \frac{1}{N_1^{\mathrm{r}}(T)} + \frac{1}{N_2^{\mathrm{r}}(T)} = \frac{1}{N_1^{\mathrm{r}}(T-1)} + \frac{1}{N_2^{\mathrm{r}}(T-1)} - V_l. \end{aligned}$$

The probabilities are taken under their respective marginal laws. The displayed lower bound g_1 is strictly positive whenever $|\hat{\Delta}^{\mathrm{r}}(T-1)| > 0$. When $\hat{\Delta}^{\mathrm{r}}(T-1) = 0$, the one-step improvement itself is still strictly positive, but the particular lower bound above is zero; the last assertion below supplies the bound used for that case. In particular, if $N_1^{\mathrm{r}}(T-1)$ and $N_2^{\mathrm{r}}(T-1)$ are positive integers smaller than T and $T^{-2} \leq |\hat{\Delta}^{\mathrm{r}}(T-1)| \leq 2T^{-2}$, then

$$g_1(\hat{\Delta}^{\mathrm{r}}(T-1), N_1^{\mathrm{r}}(T-1), N_2^{\mathrm{r}}(T-1), i) = \Omega(T^{-6}).$$

The implied constant is uniform over the stated count and gap ranges and over $i \in \{1, 2\}$. Moreover, if $N_1^{\mathrm{r}}(T-1), N_2^{\mathrm{r}}(T-1) \leq T'$ and $|\hat{\Delta}^{\mathrm{r}}(T-1)| \leq 10\sqrt{\log T}/T$, then, for either $i = 1, 2$,

$$L(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1), T+1) - L_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1), T) = \Omega(T^{-22}).$$

Proof of Lemma 15. By the update rule of the Bayesian posterior, we have

$$\hat{\Delta}^{\mathrm{r}}(T) - \hat{\Delta}^{\mathrm{r}}(T-1) \sim \mathcal{N}(0, V_l). \tag{16}$$

On the one hand,

$$L(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1), T+1) = \mathbb{E}_{\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1)}\left[\Delta_{\widehat{J}_{\{1,2\}}(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1))}\right]$$

is the loss when we make the immediate Bayes recommendation after round $T-1$. On the other hand,

$$L_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1), T)$$

is the loss when we sample $I^{\mathrm{r}}(T) = i$ and chooses recommendation arm $J^{\mathrm{r}}(T)$ after we update $\hat{\Delta}^{\mathrm{r}}(T-1)$. The sample $I^{\mathrm{r}}(T) = i$ reduces the posterior variance from $V_r + V_l$ to V_r.

For ease of discussion,[13] assume that $\hat{\Delta}^{\mathrm{r}}(T-1) \geq 0$. Let Φ_r and Φ_l be the corresponding Gaussian probability measures with variances V_r and V_l, and put $d = \hat{\Delta}^{\mathrm{r}}(T-1) \geq 0$. We have

$$\begin{aligned}
&L(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1), T+1) - L_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1), T) \\
&= \int\int ((r+l)-d)_+ d\Phi_r(r) d\Phi_l(l) \\
&\qquad - \int\int (r - |d-l|)_+ d\Phi_r(r) d\Phi_l(l) \\
&\quad \text{(by Lemma 14 and Eq. (16))} \\
&\geq \mathbb{P}\left[\frac{3}{2}d \leq l \leq 2d\right] \times \mathbb{P}[r \geq d] \times d. \qquad (17)
\end{aligned}$$

Indeed, for $l \leq d$ the two integrands coincide. For $l > d$, writing $u = l - d$, their difference is $(r+u)_+ - (r-u)_+ \geq 0$; on $\{3d/2 \leq l \leq 2d,\ r \geq d\}$ it is at least d.

We next evaluate Eq. (17) when $T^{-2} \leq d \leq 2T^{-2}$. The stated count condition implies that $V_r, V_l \geq 1/T^2$, and thus

$$\begin{aligned}
&\mathbb{P}\left[\frac{3}{2}d \leq l \leq 2d\right] \\
&:= \frac{1}{\sqrt{2\pi V_l}} \int \mathbf{1}[3d/2 \leq l \leq 2d] \exp\left(-\frac{l^2}{2V_l}\right) dl \\
&\geq \frac{1}{\sqrt{2\pi}} \int \mathbf{1}[3d/2 \leq l \leq 2d] \exp\left(-\frac{T^2 l^2}{2}\right) dl \\
&\geq \frac{1}{\sqrt{2\pi}} \times \frac{1}{2T^2} \times \exp\left(-T^2 \times \left(\frac{4}{T^2}\right)^2 \times \frac{1}{2}\right) \\
&= \Omega(T^{-2}).
\end{aligned}$$

After the last draw both counts are at most T, so $V_r \geq 2/T$. Hence, uniformly for $d \leq 2T^{-2}$,

$$\mathbb{P}[r \geq d] = \Phi^c\left(\frac{d}{\sqrt{V_r}}\right) \geq \frac{1}{4}$$

for all sufficiently large T. Together with the preceding $\Omega(T^{-2})$ interval probability and $d \geq T^{-2}$, this gives the stronger bound $\Omega(T^{-4})$, and in particular the stated $\Omega(T^{-6})$ bound.

For the last assertion, let $a = T^{-2}$ and use instead the event

$$d + a \leq l \leq d + 2a, \qquad r \geq 2a.$$

[13] Deriving the same result in the case of $\hat{\Delta}^{\mathrm{r}}(T-1) \leq 0$ is straightforward.

On this event the pointwise improvement is at least $2a$. If both current counts are at most T', then $V_l \geq 1/(T'(T'+1)) > 4/T^2$. Under $d \leq 10\sqrt{\log T}/T$, the Gaussian density of l throughout the displayed interval is at least a constant times T^{-18}. Hence the probability of the l-event is $\Omega(T^{-20})$. Also $\mathbb{P}[r \geq 2a] \geq 1/4$ for all sufficiently large T, since $V_r \geq 2/(T'+1)$. The resulting improvement is $\Omega(T^{-2}T^{-20}) = \Omega(T^{-22})$. $\square$

The following lemma states that the optimal strategy at round T is to draw the arm given by $\arg\min_i N_i^{\mathrm{r}}(T-1)$.

Lemma 16. (Optimal action at round T) Assume that $N_1^{\mathrm{r}}(T-1)$ and $N_2^{\mathrm{r}}(T-1)$ are positive integers with $N_1^{\mathrm{r}}(T-1) + N_2^{\mathrm{r}}(T-1) = 2T' - 1$. Let $i = \arg\max_{k\in\{1,2\}} N_k^{\mathrm{r}}(T-1)$ and let $j \neq i$. The odd total implies $N_j^{\mathrm{r}}(T-1) < N_i^{\mathrm{r}}(T-1)$, with $N_i^{\mathrm{r}}(T-1) \geq T'$ and $N_j^{\mathrm{r}}(T-1) \leq T'-1$. Then,

$$L_i(\boldsymbol{H}_{\setminus 3}^{\mathrm{r}}(T-1), T) - L_j(\boldsymbol{H}_{\setminus 3}^{\mathrm{r}}(T-1), T) \geq g_2(\hat{\Delta}^{\mathrm{r}}(T-1), N_1^{\mathrm{r}}(T-1), N_2^{\mathrm{r}}(T-1))$$

for some nonnegative function $g_2(\hat{\Delta}^{\mathrm{r}}(T-1), N_1^{\mathrm{r}}(T-1), N_2^{\mathrm{r}}(T-1))$; the inequality is strict whenever the two counts differ. In particular, if $T^{-2} \leq |\hat{\Delta}^{\mathrm{r}}(T-1)| \leq 2T^{-2}$, then

$$g_2(\hat{\Delta}^{\mathrm{r}}(T-1), N_1^{\mathrm{r}}(T-1), N_2^{\mathrm{r}}(T-1)) = \Omega(T^{-8}).$$

The implied constant is uniform over all states satisfying the stated count and gap conditions.

Proof of Lemma 16. Let $V(N) = 1/(N(N+1))$. By assumption, $N_i^{\mathrm{r}}(T-1) > N_j^{\mathrm{r}}(T-1)$ and thus $V(N_i^{\mathrm{r}}(T-1)) < V(N_j^{\mathrm{r}}(T-1))$. By the update rule of the Bayesian posterior, if $I^{\mathrm{r}}(T) = i$, then we have

$$\hat{\Delta}^{\mathrm{r}}(T) - \hat{\Delta}^{\mathrm{r}}(T-1) \sim \mathcal{N}\left(0, V(N_i^{\mathrm{r}}(T-1))\right).$$

Moreover, the distribution of a normal variable with variance $V(N_j^{\mathrm{r}}(T-1))$ is the same as the distribution of two independent normal variables with variances $V(N_i^{\mathrm{r}}(T-1))$ and $V(N_j^{\mathrm{r}}(T-1)) - V(N_i^{\mathrm{r}}(T-1))$, and so, if $I^{\mathrm{r}}(T) = j$, then

$$\begin{aligned}\hat{\Delta}^{\mathrm{r}}(T) - \hat{\Delta}^{\mathrm{r}}(T-1) &\sim \mathcal{N}\left(0, V(N_j^{\mathrm{r}}(T-1))\right) \\ &\sim \mathcal{N}\left(0, V(N_i^{\mathrm{r}}(T-1))\right) + \mathcal{N}\left(0, V(N_j^{\mathrm{r}}(T-1)) - V(N_i^{\mathrm{r}}(T-1))\right).\end{aligned}$$

Let

$$\begin{aligned}&r \sim \mathcal{N}(0, V_r), \text{ where } V_r = \frac{1}{N_i(T-1)} + \frac{1}{N_j(T-1)+1}, \\ &l \sim \mathcal{N}(0, V_l), \text{ where } V_l = V(N_i(T-1)), \\ &g \sim \mathcal{N}(0, V_g), \text{ where } V_g = V(N_j(T-1)) - V(N_i(T-1))\end{aligned}$$

be three independent zero-mean normal random variables. Note also that the posterior variance is

$$\frac{1}{N_1^{\mathrm{r}}(T)} + \frac{1}{N_2^{\mathrm{r}}(T)} = \begin{cases} V_r + V_g & (\text{if } I(T) = i), \\ V_r & (\text{otherwise, if } I(T) = j). \end{cases}$$

By symmetry, assume $d := |\hat{\Delta}^{\mathrm{r}}(T-1)| = \hat{\Delta}^{\mathrm{r}}(T-1) > 0$. Then,

$$\begin{aligned}
&L_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1), T) - L_j(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1), T) \\
&= \int\int\int ((r+g) - |\hat{\Delta}^{\mathrm{r}}(T-1) - l|)_+ d\Phi_r(r) d\Phi_l(l) d\Phi_g(g) \\
&\quad - \int\int\int (r - |\hat{\Delta}^{\mathrm{r}}(T-1) - l - g|)_+ d\Phi_r(r) d\Phi_l(l) d\Phi_g(g). \qquad (18)
\end{aligned}$$

For fixed l, put $\delta = d - l$ and integrate first over (r, g). The resulting conditional difference is the Bayes value of observing the additional Gaussian signal g when the current posterior-mean difference is δ. It is nonnegative: the policy that ignores g is feasible after observing it. Equivalently, this follows directly from conditional Bayes optimality. Thus the portions of the l-integral outside any selected set may be discarded after, rather than before, integration over (r, g).

On $d/4 \le l \le d/3$, we have $\delta \in [2d/3, 3d/4] > 0$. For such a fixed l, the integrand in Eq. (18) is pointwise nonnegative: it is zero when $g \le \delta$, and when $g > \delta$ it equals

$$(r + g - \delta)_+ - (r - (g - \delta))_+ \ge 0.$$

Moreover, on $3\delta/2 \le g \le 2\delta$ and $r \ge \delta$, it is at least δ. The intervals used below satisfy these conditions uniformly over $l \in [d/4, d/3]$. Therefore, Eq. (18) gives

$$\begin{aligned}
L_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1), T) - L_j(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1), T) &\ge \mathbb{P}\left[\frac{d}{4} \le l \le \frac{d}{3}\right] \times \mathbb{P}\left[\frac{9d}{8} \le g \le \frac{4d}{3}\right] \times \mathbb{P}\left[r \ge \frac{3d}{4}\right] \times \frac{2d}{3} \\
&=: g_2(\hat{\Delta}^{\mathrm{r}}(T-1), N_1^{\mathrm{r}}(T-1), N_2^{\mathrm{r}}(T-1)).
\end{aligned}$$

When $d = 0$, strict positivity follows by first conditioning on any $l \ne 0$ (an event of probability one) and applying the same calculation with $\delta = |l|$.

We next evaluate g_2 for $T^{-2} \le d \le 2T^{-2}$. Since the two counts are distinct positive integers no larger than T, the function $V(n) = 1/[n(n+1)]$ gives

$$V_l = V(N_i^{\mathrm{r}}(T-1)) \ge \frac{1}{T(T+1)} \ge T^{-3},$$

and, because $N_j^{\mathrm{r}}(T-1) \le N_i^{\mathrm{r}}(T-1) - 1$,

$$\begin{aligned}
V_g &= V(N_j^{\mathrm{r}}(T-1)) - V(N_i^{\mathrm{r}}(T-1)) \\
&\ge V(N_i^{\mathrm{r}}(T-1) - 1) - V(N_i^{\mathrm{r}}(T-1)) = \frac{2}{(N_i^{\mathrm{r}}(T-1) - 1) N_i^{\mathrm{r}}(T-1)(N_i^{\mathrm{r}}(T-1) + 1)} \ge T^{-3},
\end{aligned}$$

whereas both variances are at most $1/2$. On the two integration intervals, $|l|, |g| \le 3T^{-2}$, so the corresponding Gaussian densities are bounded below by a positive universal constant for all sufficiently large T. Both interval probabilities are therefore $\Omega(T^{-2})$. Moreover, $V_r \ge T^{-1}$ implies $\mathbb{P}[r \ge 3d/4] \ge 1/4$ for all sufficiently large T. Multiplication by $2d/3 = \Omega(T^{-2})$ yields the stronger $\Omega(T^{-6})$ bound and hence

the stated $g_2 = \Omega(T^{-8})$ bound. In particular,

$$\mathbb{P}[d/4 \le l \le d/3] = \Omega(T^{-2}). \tag{19}$$

□

E.2.2 Characterization of the two-armed problem

in Section E.2.1, the reduction of the loss by the last sample was quantified. Based on this result, the following lemma provides an explicit count-balancing Bayes optimal continuation for the two-armed problem.

Lemma 17. For every proper two-armed posterior $\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(s-1)$ with positive-integer precisions and every stage $s \in \{K+1, \dots, T\}$, every arm $i \in \arg\min_{k\in\{1,2\}} N^{\mathrm{r}}_k(s-1)$ belongs to $\arg\min_{j\in\{1,2\}} L_j(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(s-1), s)$.

Proof of Lemma 17. Write the current posterior of the difference $D = \mu_1 - \mu_2$ as

$$D \sim \mathcal{N}(d, v), \qquad v = \frac{1}{n_1} + \frac{1}{n_2},$$

where $n_k = N^{\mathrm{r}}_k(s-1)$. Consider first a deterministic allocation of the remaining $S = T - s + 1$ samples, and let m_1, m_2 be its final counts. Gaussian conjugacy gives

$$\hat{D}(T) - d \sim \mathcal{N}\left(0, v - w\right), \qquad w = \frac{1}{m_1} + \frac{1}{m_2}. \tag{20}$$

Here $\hat{D}(T) := \hat{\mu}^{\mathrm{r}}_1(T) - \hat{\mu}^{\mathrm{r}}_2(T)$ denotes the terminal posterior-mean difference under this deterministic allocation. The order of the samples does not affect this law.

Conditionally on the current posterior, the expected terminal reward of the recommended arm is

$$\begin{aligned}\mathbb{E}[\mu_{J(T)}] &= \mathbb{E}\left[\max_{k=1,2} \mathbb{E}[\mu_k \mid \boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T)]\right] \\ &= \frac{1}{2}\mathbb{E}\left[\hat{\mu}^{\mathrm{r}}_1(T) + \hat{\mu}^{\mathrm{r}}_2(T) + |\hat{D}(T)|\right].\end{aligned}$$

The quantity $\mathbb{E}[\max(\mu_1, \mu_2)]$ under the current posterior is fixed before the experiment, so minimizing posterior expected simple regret is equivalent to maximizing this expected terminal reward. The expectation of the sum of the two posterior means is fixed by the martingale property. Moreover, for fixed d, the quantity

$$\mathbb{E}\left[|d + \sqrt{q}Z|\right], \qquad Z \sim \mathcal{N}(0,1),$$

is nondecreasing in q, and is strictly increasing whenever the two compared values of q differ. Indeed, if $q_2 > q_1$, write a $\mathcal{N}(0, q_2)$ variable as the sum of independent $\mathcal{N}(0, q_1)$ and $\mathcal{N}(0, q_2 - q_1)$ variables and apply conditional Jensen's inequality to the convex function $x \mapsto |x|$; strictness follows because the added Gaussian variable has positive variance. It follows from Eq. (20) that a deterministic allocation minimizes the terminal simple regret exactly when it minimizes $w = 1/m_1 + 1/m_2$.

Among all integer pairs satisfying $m_k \ge n_k$ and $m_1 + m_2 = n_1 + n_2 + S$, this convex objective is minimized by making the final counts as equal as the lower bounds $m_k \ge n_k$ permit. Drawing an arm

with the smaller current count and repeating this rule produces precisely such a pair.

It remains to justify that an adaptive allocation cannot do better. We use backward induction on S. The assertion for $S = 1$ follows from the preceding comparison. Assuming it for $S - 1$, after either first action an optimal continuation can be chosen to follow the smaller-count rule and hence is deterministic as a function of the counts. The two possible first actions therefore reduce to the deterministic final-count comparison above. An action on a smaller-count arm leads to a minimizing final pair, so it minimizes $L_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(s-1), s)$. If both first actions lead to the same optimal final pair up to order, both are optimal; otherwise drawing the smaller-count arm is strictly better. □

E.2.3 A lower bound for the EBI in the two-armed problem

Section E.1 reduced the three-armed instance with posterior $\boldsymbol{H}^{\mathrm{r}}(t-1)$ to the two-armed instance with posterior $\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1)$. Section E.2.2 showed that the two-armed problem admits a Bayes optimal continuation following the smaller-count rule. Given these results, we bound the EBI from below in the two-armed best arm identification problem. Assuming event $\mathcal{W}(t)$, the following lemma provides a lower bound for the probability that $\hat{\Delta}^{\mathrm{r}}(T-1)$ lies in a small region around zero.

Lemma 18. Assume $t < T$. Fix a deterministic tie-breaking rule that depends only on the two current draw counts, and suppose that, for $s = t, t+1, \ldots, T-1$,

$$I^{\mathrm{r}}(s) \in \operatorname*{arg\,min}_{i \in \{1,2\}} N_i^{\mathrm{r}}(s-1),$$

with ties resolved by this fixed rule. Let $\boldsymbol{H}^{\mathrm{r}}(t-1) = \boldsymbol{H}(t-1)$ be any posterior such that $\mathcal{W}(t)$ holds, and let $\hat{\Delta}(t-1)$, $N_1(t-1)$, and $N_2(t-1)$ be the statistics determined by $\boldsymbol{H}(t-1)$. Let $\mathbb{P}^{\mathrm{bal}}_{\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1)}$ denote the posterior-predictive law of the virtual rewards and posterior updates, conditional on the current reduced posterior, under this fixed smaller-count rule. This is distinct from the frequentist reward-stack law used to define $\mathcal{W}(t)$. Then,

$$\mathbb{P}^{\mathrm{bal}}_{\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1)}\left[\frac{1}{T^2} \le \hat{\Delta}^{\mathrm{r}}(T-1) \le \frac{2}{T^2}\right] \ge c_3 T^{-170}$$

for a constant $c_3 > 0$ independent of t, T, the realized posterior, and the fixed count-only tie-breaking rule, for all sufficiently large T.

Proof of Lemma 18. Put $n_i = N_i(t-1)$, and let $m_i = N_i^{\mathrm{r}}(T-1)$ be the counts produced by the smaller-count rule. The rule depends only on the counts, so m_i is deterministic given (n_1, n_2). By Gaussian conjugacy,

$$\hat{\Delta}^{\mathrm{r}}(T-1) - \hat{\Delta}(t-1) \sim \mathcal{N}(0, V_s), \qquad V_s = \sum_{i=1}^{2}\left(\frac{1}{n_i} - \frac{1}{m_i}\right). \tag{21}$$

Because $t < T$, at least one virtual draw occurs before round T. Each such draw, when the sampled arm has count q, contributes $1/[q(q+1)]$ to V_s. Under $\mathcal{W}(t)$ and the smaller-count rule, the terminal counts at time $T-1$ are T' and $T'-1$, so $q+1 \le T'$ and hence

$$V_s \ge \frac{1}{T'(T'-1)} \ge T^{-2}$$

for all sufficiently large T.

Define $v_i^+ = 1/n_i - 1/T'$. Since $\{m_1, m_2\} = \{T', T'-1\}$,

$$v_1^+ + v_2^+ = V_s + \frac{1}{T'-1} - \frac{1}{T'} = V_s + \frac{1}{T'(T'-1)}.$$

The preceding lower bound on V_s therefore gives

$$v_1^+ + v_2^+ \le 2V_s. \tag{22}$$

Using $\mathcal{Y}_i$, $\mathcal{Z}_i$, and $m^{-2} \le 2/[m(m+1)]$, we obtain

$$\begin{aligned} |\hat{\Delta}(t-1)| &\le 3\sqrt{\log T}\sum_{i=1}^{2}\sqrt{\sum_{m=n_i}^{T'-1}\frac{1}{m^2} + \frac{2}{T^2}}, \\ (\hat{\Delta}(t-1))^2 &\le 160(\log T)V_s. \end{aligned} \tag{23}$$

Here the second line follows from Eq. (22), $(\sqrt{x}+\sqrt{y})^2 \le 2(x+y)$, and $V_s \ge T^{-2}$.

Let $a = T^{-2}$. From Eq. (21), the probability in the lemma equals

$$\frac{1}{\sqrt{2\pi V_s}}\int_{a-\hat{\Delta}(t-1)}^{2a-\hat{\Delta}(t-1)} \exp\left(-\frac{z^2}{2V_s}\right) dz.$$

We have $V_s \le 2$, so the normalizing factor is bounded below by a positive constant. Throughout the integration interval, Eqs. (23) and $V_s \ge T^{-2}$ give

$$\frac{z^2}{2V_s} \le \frac{(|\hat{\Delta}(t-1)| + 2a)^2}{2V_s} \le 164\log T$$

for all sufficiently large T. Since the interval has length a, the probability is at least cT^{-166} for a universal $c > 0$. Decreasing c_3 if necessary, this is at least $c_3 T^{-170}$ for all sufficiently large T. □

Based on the previous lemmas, we can bound $B_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t)$.

Lemma 19. Assume that $t < T$ and $\mathcal{W}(t)$ holds. There are constants $c_1, c_2 > 0$, independent of t, T, and the realized posterior, such that, for all sufficiently large T and each $i \in \{1, 2\}$,

$$B_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t) \ge c_1 T^{-176}. \tag{24}$$

Moreover, if $N_i(t-1) < T'$ and $N_j(t-1) \ge T'$, then

$$B_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t) - B_j(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t) \ge c_2 T^{-178}. \tag{25}$$

Proof. We first derive Eq. (24). By definition,

$$B_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t) := L(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t+1) - L_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t).$$

Lemma 17 allows us to choose, in each of the two value functions, an optimal continuation that always samples a smaller-count arm and resolves equal counts by the same fixed deterministic count-only rule. Here is the count argument that permits the common-sample coupling. Under $\mathcal{W}(t)$, $N_3(t-1)=1$, and hence $N_1(t-1)+N_2(t-1)=t-2$. The continuation defining $L(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t+1)$ has $T-t$ samples, so its terminal count total is $T-2=2T'-1$; a balanced optimal allocation has terminal count multiset $\{T', T'-1\}$. If the forced arm i currently has count below T', choose the orientation in which its skip-side terminal count is $T'-1$; the forced problem then ends at (T', T'). If its current count is already T', the forced problem instead ends with count multiset $\{T'+1, T'-1\}$. Thus, in both cases, an optimal forced allocation is an optimal skip allocation plus one sample from arm i.

For a deterministic allocation, Eq. (20) and the martingale argument following it show that the posterior-predictive terminal risk depends on the allocation only through its terminal counts, and not on the order of its samples. Consequently, although $L_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t)$ requires the arm-i sample at stage t, for the purpose of evaluating its risk we may couple the observations common to the two allocations, place those observations first in the same deterministic order, and place the additional arm-i observation last.

Immediately before this additional observation, the counts are T' and $T'-1$. Conditional on the common observations, the difference between the two losses is therefore the one-step improvement due to this last sample. Lemma 18 gives a uniform $c_3 T^{-170}$ lower bound on the displayed penultimate event below, and Lemma 15 gives a uniform $c_4 T^{-6}$ lower bound on the one-step improvement on that event. Therefore,

$$
\begin{aligned}
&L(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t+1) - L_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t) \\
&\geq \mathbb{P}^{\mathrm{bal}}_{\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1)}\left[\frac{1}{T^2} \leq \hat{\Delta}^{\mathrm{r}}(T-1) \leq \frac{2}{T^2}\right] \times c_4 T^{-6} \\
&\quad \text{(by Lemma 15 and the discussion above)} \\
&\geq c_3 c_4 T^{-176}. \\
&\quad \text{(by Lemma 18)}
\end{aligned}
$$

Taking $c_1 = c_3 c_4$ proves Eq. (24).

We next derive Eq. (25). Notice that under $N_j(t-1) \geq T'$ and $N_i(t-1) < T'$, drawing the more-sampled arm j results in final counts $T'+1$ and $T'-1$, which is suboptimal in the two-armed problem. By Lemma 17, all samples common to the two allocations can be placed first, in any order, and are allocated to the under-sampled arm i. The two experiments then differ only in their last sample: the balanced allocation samples i, whereas the unbalanced allocation samples j. The common penultimate counts are again T' and $T'-1$. On the same event, Lemma 16 gives a uniform $c_5 T^{-8}$ advantage for sampling the smaller-count arm, while Lemma 18 supplies the factor $c_3 T^{-170}$. From the discussion above, we have

$$
\begin{aligned}
&L_j(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t) - L_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t) \\
&\geq \mathbb{P}^{\mathrm{bal}}_{\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1)}\left[\frac{1}{T^2} \leq \hat{\Delta}^{\mathrm{r}}(T-1) \leq \frac{2}{T^2}\right] \times c_5 T^{-8}
\end{aligned}
$$

(by Lemma 16 and the discussion above)

$\geq c_3 c_5 T^{-178}$.

(by Lemma 18)

Taking $c_2 = c_3 c_5$ proves Eq. (25). □

E.3 Proof of Lemma 7

Proof of Lemma 7. For $i \in \{1, 2\}$, we have

$$\begin{aligned}
B_i(\boldsymbol{H}^{\mathrm{r}}(t-1), t) &:= L(\boldsymbol{H}^{\mathrm{r}}(t-1), t+1) - L_i(\boldsymbol{H}^{\mathrm{r}}(t-1), t) \\
&\geq L(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t+1) - L_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t) - 2\mathrm{R}^{(3)}_{\boldsymbol{H}^{\mathrm{r}}(t-1)} \\
&\quad \text{(by Lemma 13)} \\
&\geq c_1 T^{-176} - 2\mathrm{R}^{(3)}_{\boldsymbol{H}^{\mathrm{r}}(t-1)} \\
&\quad \text{(by Lemmas 11 and 19)} \\
&\geq c_1 T^{-176} - 2g_0 \\
&= \Omega(T^{-176}) - o(T^{-180}) = \Omega(T^{-176}) = \Omega(T^{-178}).
\end{aligned}$$

Moreover, let $N_j(t-1) = T'$ and $i \neq j$. Then $N_i(t-1) < T'$, and

$$\begin{aligned}
&B_i(\boldsymbol{H}^{\mathrm{r}}(t-1), t) - B_j(\boldsymbol{H}^{\mathrm{r}}(t-1), t) \\
&\geq B_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t) - B_j(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(t-1), t) - 4\mathrm{R}^{(3)}_{\boldsymbol{H}^{\mathrm{r}}(t-1)} \\
&\quad \text{(by Lemma 13)} \\
&\geq c_2 T^{-178} - 4g_0 \\
&\quad \text{(by Lemmas 11 and 19)} \\
&= \Omega(T^{-178}) - o(T^{-180}) = \Omega(T^{-178}).
\end{aligned}$$

□

Proof of Lemma 8. Under $\mathcal{W}(T)$, $N_3(T-1) = 1$ and $N_1(T-1) + N_2(T-1) = T - 2 = 2T' - 1$. Since both of the latter counts are at most T', they are T' and $T' - 1$. Events $\mathcal{Y}_i$ and $\mathcal{Z}_i$ imply

$$|\hat{\mu}_1(T-1) - \hat{\mu}_2(T-1)| \leq \frac{2}{T^2} + \frac{3\sqrt{\log T}}{T' - 1} \leq \frac{10\sqrt{\log T}}{T}$$

for all sufficiently large T. The last assertion of Lemma 15 therefore gives, for $i = 1, 2$,

$$B_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1), T) = \Omega(T^{-22}).$$

Lemmas 13 and 11 now yield

$$B_i(\boldsymbol{H}^{\mathrm{r}}(T-1),T) \geq B_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1),T) - 2g_0 = \Omega(T^{-22}).$$

On the other hand, Lemma 6 gives $B_3(\boldsymbol{H}^{\mathrm{r}}(T-1),T) = o(T^{-180})$. Hence arm 3 cannot maximize the EBI in the last round. □

F Discussion on the Tightness of the Bound

The goal of Theorem 2 was to derive a polynomial simple regret lower bound, and its polynomial exponent is not expected to be tight. The following observation suggests how the probability estimate could be improved, but it is not used in the theorem and is not asserted here as a separate proved bound. One may attempt to replace

$$\mathcal{Y}_i := \left\{ |\hat{\mu}_{i,T'}| \leq \frac{1}{T^2} \right\}$$

with a slightly larger bound of

$$\left\{ |\hat{\mu}_{i,T'}| \leq \frac{1}{T^{3/2}} \right\}.$$

The limiting calculation is Eq. (19), in which the exponent of $\exp\left(-T^3 l^2/2\right)$ remains $O(1)$ at the $T^{-3/2}$ scale.[14] A formal improvement would require rechecking all uniform bounds in Lemmas 6–8 with the enlarged event. We retain the proved T^{-2} event for ease of exposition.

[14] This part bounds $|L_i(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1),T) - L_j(\boldsymbol{H}^{\mathrm{r}}_{\setminus 3}(T-1),T)|$ from below. This value is decomposed into several components of poly(T^{-1}). In particular, Eq. (19) corresponds to the probability that a normal random variable with variance T^{-3} is larger than a given $\hat{\mu}^{\mathrm{r}}_i(T-1) - \hat{\mu}^{\mathrm{r}}_j(T-1)$. For this probability to be poly(T^{-1}), we require $|\hat{\mu}^{\mathrm{r}}_i(T-1) - \hat{\mu}^{\mathrm{r}}_j(T-1)| = O(T^{-3/2})$.